The Three-Level Posterior Tower: Hierarchical Bayesian Inference and Mixed Distributive Laws

Revised version 10

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Abstract

We construct a categorical framework for structured Bayesian inference that keeps three levels of posterior information as state: a posterior ρ over outcomes, a second-order posterior Π over outcome posteriors, a third-order posterior Ω, and a shadow Σ over structured hypotheses. Combining this state with the finite distribution monad D forces a design choice. In the *absolute* tower $\mathsf{C}_3(X) = X \times \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathrm{Sh}(X)$ the levels are re-indexed by the object, so $\mathsf{C}_3(\mathsf{D}X)$ already contains D^4X: the hierarchy grows by one order every time the monad is applied. We prove that $X \times \mathsf{E}(X)$ is a comonad for every endofunctor E, that the context and distribution-level lifts are coherent by functoriality, and that the two natural candidate laws $\mathsf{DC}_3 \Rightarrow \mathsf{C}_3\mathsf{D}$ each fail at least one Beck axiom; the unit axiom alone forbids point-mass hierarchy at level one. We then construct a *base-relative* tower, in which the three levels live over a fixed outcome base B and do not grow. Its *coherent* contexts, those whose barycentric projections and shadow predictive agree, form a sub-algebra of the monad, and the canonical hierarchy ρ_h, $\Pi_h = \delta_{\rho_h}$, $\Omega_h = \delta_{\Pi_h}$ extends to an affine section of it. For this tower we obtain a strict mixed distributive law satisfying all four Beck axioms, natural in the base, and the whole construction (coherence, canonical section, law, admissibility data and obstructions) is transported along stochastic changes of base, that is along Markov kernels $B \to \mathsf{D}(B')$, where obstructions can disappear but never appear and certificates pull back. Pooling commutes with hierarchy formation, conditioning commutes with pooling after evidence reweighting, and distinct posteriors are retained at level two. We then pose a coherent lifting problem: given admissible first- and second-order states generated by a finite hypothesis class, when does a claimed posterior state extend to a full coherent three-level context? Admissibility at each level is a convex condition, failure is monotone up the tower, each level can be the first to fail, and every failure carries an explicit certificate: a separating test function at level one (with the obstruction size given by a minimax identity) and a witness state at levels two and three. Pooling can hide level-one obstructions but never level-two or level-three ones. When the admissible second-order states are restricted, a claim can pass every local test and still admit no coherent lift; we characterize such claims by effective admissible sets with explicit certificates, including second-order test functions on posteriors, and define a secondary obstruction on the fibre of level-two lifts whose value depends on the chosen lift and ranges over an interval that contains zero exactly when a lift exists. This gives a stratified, precise candidate definition of a hallucinated claim as one whose required coherent lift is obstructed. We also characterize when a mixed context can be attributed to its sources: weights are unique exactly for affinely independent sources, the supports-only case gives a sharp

*Code and verification scripts: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/Heirarchical_posteriors

belief–plausibility interval, and under contamination by unmodeled sources the data certify a lower bound on the unmodeled weight, while every exclusion depends on an assumed budget and the number of unmodeled sources is not identifiable. Analytic sieving and its parity barrier serve as a classical illustration of iterated conditioning and observational equivalence. We identify the coalgebras of the base-relative comonad with sets over the context algebra and its bialgebras with convex spaces over it, and we make precise that the ambient 2-category is **Cat**, in which these categories are locally discrete, so two-dimensional monad theory adds no content here. We then test the natural relaxation of the absolute tower by the convex order: the unit and multiplication defects of the shift law are comparable but in opposite orientations relative to the Beck diagrams, and the comultiplication defect is incomparable, for both candidate laws. The naive convex-order relaxation therefore fails, which motivates an enrichment that compares coherent lifts rather than barycentric spreads. On the algebraic side we give a normalized partial composition on structured posteriors, corrected conditioning laws, an exact characterization of when observational equivalence is a congruence, and a proof that the likelihood-factorization "iff" holds under nondegeneracy alone. All identities and counterexamples were checked in exact rational arithmetic; the scripts are in the accompanying repository.

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1 Introduction

Bayesian inference in structured problems produces a hierarchy of posteriors: a posterior over outcomes, a posterior over hypotheses (each of which itself predicts outcomes), and a posterior over the ways hypotheses combine. This paper builds a categorical framework that retains all three levels as state rather than as intermediate computations, and asks how such state interacts with the probability monad.

The framework builds on the three-layer architecture of [9]; the hierarchy layer is developed in [7] and the holonomy obstruction in [8].

The design choice. The levels of the tower are indexed by the object on which the comonad is evaluated. Evaluating at $\mathsf{D}X$, as any distributive law $\mathsf{D}\mathsf{C}_3 \Rightarrow \mathsf{C}_3\mathsf{D}$ must, shifts every level up by one order:

$$\mathsf{C}_3(\mathsf{D}X) = \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{D}^4X \times \mathrm{Sh}(\mathsf{D}X).$$

There are two ways to proceed. *Option 1* keeps this absolute indexing and accepts a target that contains a fourth-order level (section 8); the transformation of the first version omitted that level and was therefore ill-typed. *Option 2* fixes an outcome base B once and for all and lets the three levels live over B whatever object the comonad is evaluated at (section 9). Option 2 preserves exactly three hierarchical levels and is the main construction of this paper; it admits a strict mixed distributive law (theorem 9.8). Option 1 is developed to show precisely what goes wrong without a fixed base.

Contributions.

1. A base-relative three-level hierarchy over a fixed base B, with coherent contexts, the canonical hierarchy $\sigma(\delta_h) = (\rho_h, \delta_{\rho_h}, \delta_{\delta_{\rho_h}}, \delta_h)$ as an affine section, and a strict mixed distributive law satisfying all four Beck axioms, natural in the base (section 9); the construction is functorial along Markov kernels, not only functions (propositions 9.14 and 9.16 and corollary 9.15).

2. A coherent lifting problem for claims against the tower, with level-wise obstruction certificates, strictness and monotonicity of failure levels, and the behavior of obstructions under pooling (section 10); and an obstruction tower with indeterminacy, in which a secondary obstruction, defined only once the primary one vanishes, depends on the choice of intermediate lift (section 10.4).

3. The coalgebras, lifted monad and bialgebras of the base-relative law, a precise statement of the ambient 2-category and of what two-dimensional monad theory does and does not add, and convex-order Beck comparisons for the absolute tower showing that the naive convex-order relaxation fails, through an orientation obstruction and a comparability obstruction (section 13).

4. A general *context comonad* $\mathsf{C}_\mathsf{E}(X) = X \times \mathsf{E}(X)$ for any endofunctor E of **Set** (theorem 6.1), specialized to the three-level tower C_3.

5. A hypothesis functor (Hyp, q) with q natural, so that the compatibility square is naturality (section 3); the context lift $J = \mathrm{Hyp}(j)$ and the distribution-level lift K satisfy all coherence conditions by proof (propositions 6.7 and 7.2).

6. A normalized partial algebra $(\oplus, \otimes, \mathrm{Cond})$ on structured posteriors with corrected laws (section 4).

7. An affine first isomorphism theorem for observational equivalence, and an exact characterization of when it is a congruence for $\otimes$ and for Cond (section 5); identifiability of source weights, a sharp supports-only interval, and robust decomposition under unmodeled contamination with a certified lower bound on the unmodeled weight (sections 5.2 and 5.3).

8. For the absolute tower, two mixed distributive laws λ^{σ} and $\bar{\lambda}$ with a complete account of which Beck axioms each satisfies (section 8, summarized in table 1), together with a structural lemma showing that the unit axiom forbids point-mass hierarchy at level one (lemma 8.1).

9. A proof that composite identifiability trivializes the hypothesis space, and a proof that the likelihood-factorization "iff" holds under nondegeneracy alone (remark 12.2 and theorem 12.3).

Structure of the argument. The categorical results form a progression:

(1) a strict mixed distributive law for the absolute tower fails (section 8);

(2) the base-relative reformulation restores a strict law satisfying all four Beck axioms (theorem 9.8), which remains the control result of the paper;

(3) one may ask whether an order enrichment recovers a weaker law for the absolute tower; for the convex order the answer is no, because of an orientation obstruction and a comparability obstruction (propositions 13.8 to 13.10);

(4) this leaves open which richer enrichment, comparing coherent lifts (section 10) and not only barycentric spreads, could do so (remark 13.12).

Relation to earlier versions. The first version claimed that a three-level transformation $\lambda^{[3]}$ satisfies the unit, counit and comultiplication axioms exactly and extends to a weak distributive law with a barycentric 2-cell. That transformation omitted the D^4X component of its target, and the claims do not hold; the corrected statements for the absolute tower are theorems 8.3, 8.6 and 8.7. The second version corrected the typing but left the existence of a full law open. Version 3 resolved the question for the base-relative tower; version 4 added the lifting problem; version 5 corrects the 2-categorical discussion and several empirical attributions. A full list of changes is in appendix B.

2 Preliminaries

2.1 The finite distribution monad

Let D be the finitely supported distribution monad on **Set**: $\mathsf{D}(X)$ is the set of finitely supported probability measures on X, $\mathsf{D}(f) = f_*$ is pushforward, $\eta_X(x) = \delta_x$, and $\mu_X \colon \mathsf{D}^2X \to \mathsf{D}X$ is barycenter, $\mu_X(\sum_i a_i\delta_{\rho_i}) = \sum_i a_i\rho_i$. Its Kleisli category restricted to finite sets is the category of finite stochastic matrices; see [18, 26] for the Markov-category perspective.

We work on **Set** rather than on finite sets: even for finite X with $|X| \geq 2$, the set $\mathsf{D}(X)$ is an infinite simplex, so D does not restrict to an endofunctor of finite sets. Finiteness enters only through finite supports.

Each $\mathsf{D}(Y)$ is a convex set, and every $\mathsf{D}(f)$ is affine. A product of convex sets is convex componentwise, and a product of affine maps is affine.

2.2 Mixed distributive laws

Definition 2.1 (Mixed distributive law [15, 38])**.** Let (T, η, μ) be a monad and (C, ε, δ) a comonad on the same category. A *mixed distributive law* is a natural transformation $\lambda\colon TC \Rightarrow CT$ such that

$$\begin{aligned} &\text{(U)} \quad \lambda \circ \eta_C = C\eta, && \text{(M)} \quad \lambda \circ \mu_C = C\mu \circ \lambda_T \circ T\lambda, \\ &\text{(Co)} \quad \varepsilon_T \circ \lambda = T\varepsilon, && \text{(Cm)} \quad \delta_T \circ \lambda = C\lambda \circ \lambda_C \circ T\delta. \end{aligned}$$

Weak distributive laws between two monads, in the sense of Street and Böhm [44, 16, 17], keep the multiplication-type axioms and relax the unit-type ones; Garner [28] and Goy–Petrişan [29] give probabilistic and topological examples. We return to this comparison in remark 8.9.

3 Structured hypotheses

Definition 3.1 (Hypothesis functor)**.** A *hypothesis functor* is a functor $\mathrm{Hyp}\colon \mathbf{Set} \to \mathbf{Set}$ with a natural transformation $q\colon \mathrm{Hyp} \Rightarrow \mathsf{D}$, the *predictive map*. Naturality of q is the compatibility square

$$\begin{array}{ccc} \mathrm{Hyp}(X) & \xrightarrow{q_X} & \mathsf{D}(X) \\ {\scriptstyle \mathrm{Hyp}(f)}\downarrow & & \downarrow{\scriptstyle f_*} \\ \mathrm{Hyp}(Y) & \xrightarrow[q_Y]{} & \mathsf{D}(Y) \end{array} \qquad \text{i.e.} \qquad q_Y \circ \mathrm{Hyp}(f) = f_* \circ q_X.$$

Definition 3.2 (Structured hypotheses)**.** Fix a set Θ of *structural descriptors*

$$\theta = (I_\theta, F_\theta, E_\theta, L_\theta, G_\theta, C_\theta) :$$

a sparse multi-index of directions, a factor graph, an evidence set, spline links, geometric data and conditional structure. Put

$$\mathrm{Hyp}(X) = \Theta \times \mathsf{D}(X), \qquad \mathrm{Hyp}(f)(\theta, \rho) = (\theta, f_*\rho), \qquad q_X(\theta, \rho) = \rho.$$

Lemma 3.3. Hyp *is a functor and q is natural.*

Proof. $\mathrm{Hyp} = \Theta \times \mathsf{D}(-)$ is a product of a constant functor with D, and q is the projection, which is natural because $q_Y(\mathrm{Hyp}(f)(\theta, \rho)) = f_*\rho = f_*(q_X(\theta, \rho))$. □

Remark 3.4 (Why descriptors are not transported). Directions v_j and spline links s_j are functions *on* X. Such data pull back along maps *into* X; they cannot be pushed along an arbitrary $f\colon X \to Y$ (the expression $v_j \circ f$ is not defined), and conditional structure has no canonical pushforward along non-injective maps. In CausalIDView the descriptors parametrize a model on a fixed design, and the only datum that must move along outcome maps is the predictive distribution. Definition 3.2 records exactly this. The *admissible* hypotheses $\mathrm{Hyp}^{\mathrm{adm}}(X) = \{(\theta, \rho_\theta)\}$, in which the predictive is the one produced by θ, form a subset that is generally not closed under $\mathrm{Hyp}(f)$; all constructions below are carried out on the full functor Hyp.

Definition 3.5 (Shadow)**.** $\mathrm{Sh}(X) = \mathsf{D}(\mathrm{Hyp}(X))$, the finite convex combinations $\Sigma = \sum_i \lambda_i \delta_{h_i}$ of hypotheses. Sh is a functor with $\mathrm{Sh}(f) = \mathsf{D}(\mathrm{Hyp}(f))$.

4 The structured posterior

Definition 4.1. $\mathrm{Post}(X) = \mathsf{D}(X \times \mathrm{Hyp}(X))$. An element is $P = \sum_k \mu_k \delta_{(x_k, h_k)}$. The *outcome marginal* is $\pi_{X*} = \mathsf{D}(\mathrm{pr}_1)\colon \mathrm{Post}(X) \to \mathsf{D}(X)$, and the *predictive* is $p_X = \mu_X \circ \mathsf{D}(q_X \circ \mathrm{pr}_2)$, so $p_X(P) = \sum_k \mu_k\, q_X(h_k)$.

For a nonzero finitely supported measure $m \geq 0$ write $\mathcal{N}(m) = m/\|m\|$ for its normalization. $\mathcal{N}$ absorbs positive scalars: $\mathcal{N}(cm) = \mathcal{N}(m)$ for $c > 0$.

4.1 Mixture

Definition 4.2. $P_1 \oplus_\lambda P_2 = \lambda P_1 + (1-\lambda)P_2$ for $\lambda \in [0,1]$.

Lemma 4.3 (Barycentric associativity)**.** *For $\lambda, \mu \in [0,1]$ with $\lambda\mu < 1$ and $\nu = \frac{(1-\lambda)\mu}{1-\lambda\mu}$,*

$$(P_1 \oplus_\lambda P_2) \oplus_\mu P_3 = P_1 \oplus_{\lambda\mu} (P_2 \oplus_\nu P_3).$$

If $\lambda = \mu = 1$ both sides equal P_1 for any choice of the inner weight.

Proof. The left side is $\lambda\mu P_1 + (1-\lambda)\mu P_2 + (1-\mu)P_3$. On the right, $(1-\lambda\mu)\nu = (1-\lambda)\mu$ and $(1-\lambda\mu)(1-\nu) = 1 - \lambda\mu - (1-\lambda)\mu = 1-\mu$. The boundary case is immediate. □

4.2 Composition

Definition 4.4 (Compositional structure)**.** A compositional structure on $\mathrm{Hyp}(X)$ consists of a relation $\perp$ (compatibility), a map $\otimes$ defined on compatible pairs, and an element h_{id}, such that:

(A1) (ternary admissibility and associativity) $h_1 \perp h_2$ and $(h_1 \otimes h_2) \perp h_3$ if and only if $h_2 \perp h_3$ and $h_1 \perp (h_2 \otimes h_3)$, and then $(h_1 \otimes h_2) \otimes h_3 = h_1 \otimes (h_2 \otimes h_3)$;

(A2) (identity) $h_{\mathrm{id}} \perp h$, $h \perp h_{\mathrm{id}}$ and $h_{\mathrm{id}} \otimes h = h \otimes h_{\mathrm{id}} = h$ for all h.

Definition 4.5 (Composition of posteriors)**.** The *unnormalized product* is

$$(P_1 \circledast P_2)(x,h) = \sum_{\substack{h_1 \perp h_2 \\ h_1 \otimes h_2 = h}} P_1(x,h_1)\, P_2(x,h_2),$$

a finitely supported measure of total mass $Z(P_1,P_2) \in [0,1]$. If $Z(P_1,P_2) > 0$ put $P_1 \otimes P_2 = \mathcal{N}(P_1 \circledast P_2) \in \mathrm{Post}(X)$; otherwise $P_1 \otimes P_2$ is undefined.

Remark 4.6. Without normalization the product is not a probability distribution: if $\perp$ is total its mass is $\sum_x m_1(x) m_2(x)$, where $m_i = \pi_{X*}(P_i)$, which is strictly less than 1 unless the marginals are equal point masses.

Lemma 4.7. *$\circledast$ extends bilinearly to finitely supported measures and is associative. Consequently $(P_1 \otimes P_2) \otimes P_3 = P_1 \otimes (P_2 \otimes P_3)$, and either side is defined exactly when $P_1 \circledast P_2 \circledast P_3 \neq 0$.*

Proof. Bilinearity is clear from the formula. Expanding both triple products gives sums over triples with $(h_1 \otimes h_2) \otimes h_3 = h$, respectively $h_1 \otimes (h_2 \otimes h_3) = h$, over the admissible triples; by (A1) these index sets and terms coincide. Since $\circledast$ is bilinear and $\mathcal{N}$ absorbs positive scalars, $\mathcal{N}(\mathcal{N}(m) \circledast m') = \mathcal{N}(m \circledast m')$, so both normalized products equal $\mathcal{N}(P_1 \circledast P_2 \circledast P_3)$. If $P_1 \circledast P_2 = 0$ the triple product is 0, which gives the definedness statement. □

Proposition 4.8 (Distributivity)**.** *Let $Z_{i3} = Z(P_i, P_3)$ and suppose $Z_{13}, Z_{23} > 0$. Then*

$$(P_1 \oplus_\lambda P_2) \otimes P_3 = (P_1 \otimes P_3) \oplus_{\lambda''} (P_2 \otimes P_3), \qquad \lambda'' = \frac{\lambda Z_{13}}{\lambda Z_{13} + (1-\lambda) Z_{23}}.$$

Proof. By bilinearity $(\lambda P_1 + (1-\lambda)P_2) \circledast P_3 = \lambda Z_{13}(P_1 \otimes P_3) + (1-\lambda) Z_{23}(P_2 \otimes P_3)$; normalize. □

The weight is $\lambda'' = \lambda$ only when $Z_{13} = Z_{23}$ (or $\lambda \in \{0, 1\}$).

4.3 Conditioning

Definition 4.9. For evidence e with likelihood $L(e \mid \cdot)\colon \mathrm{Hyp}(X) \to [0, \infty)$, put $Z_e(P) = \sum_{(x,h)} L(e \mid h) P(x, h)$ and, when $Z_e(P) > 0$, $\mathrm{Cond}_e(P) = \mathcal{N}\big(L(e \mid \cdot)\, P\big)$.

Proposition 4.10 (Chain rule)**.** *Suppose $L(e_1 \wedge e_2 \mid h) = L(e_1 \mid h)\, L(e_2 \mid h, e_1)$, and write $\mathrm{Cond}_{e_2 \mid e_1}$ for conditioning with likelihood $L(e_2 \mid \cdot, e_1)$. Then $\mathrm{Cond}_{e_2 \mid e_1}(\mathrm{Cond}_{e_1} P) = \mathrm{Cond}_{e_1 \wedge e_2}(P)$ whenever the right side is defined. In particular, if e_1, e_2 are conditionally independent given h, then $\mathrm{Cond}_{e_2} \circ \mathrm{Cond}_{e_1} = \mathrm{Cond}_{e_1 \wedge e_2}$.*

Proof. $\mathcal{N}(L_2 \cdot \mathcal{N}(L_1 P)) = \mathcal{N}(L_2 L_1 P)$. □

Proposition 4.11 (Conditioning and mixture)**.** *If $Z_1 = Z_e(P_1) > 0$ and $Z_2 = Z_e(P_2) > 0$, then $\mathrm{Cond}_e(P_1 \oplus_\lambda P_2) = \mathrm{Cond}_e(P_1) \oplus_{\lambda'} \mathrm{Cond}_e(P_2)$ with $\lambda' = \lambda Z_1 / (\lambda Z_1 + (1-\lambda) Z_2)$.*

Proof. $L \cdot (\lambda P_1 + (1-\lambda) P_2) = \lambda Z_1 \mathrm{Cond}_e(P_1) + (1-\lambda) Z_2 \mathrm{Cond}_e(P_2)$; normalize. □

Proposition 4.12 (Conditioning and composition)**.** *If $L(e \mid h_1 \otimes h_2) = L(e \mid h_1) L(e \mid h_2)$ for all compatible pairs, then $\mathrm{Cond}_e(P_1 \otimes P_2) = \mathrm{Cond}_e(P_1) \otimes \mathrm{Cond}_e(P_2)$, and the left side is defined exactly when the right side is.*

Proof. Pointwise, $L(e \mid h_1 \otimes h_2) P_1(x, h_1) P_2(x, h_2) = (L P_1)(x, h_1)(L P_2)(x, h_2)$, so $L \cdot (P_1 \circledast P_2) = (L P_1) \circledast (L P_2)$. Using that $\mathcal{N}$ absorbs positive scalars and $\circledast$ is bilinear, $\mathrm{Cond}_e(P_1 \otimes P_2) = \mathcal{N}(L \cdot (P_1 \circledast P_2)) = \mathcal{N}((L P_1) \circledast (L P_2)) = \mathcal{N}(\mathrm{Cond}_e P_1 \circledast \mathrm{Cond}_e P_2)$. Both sides are defined iff $(L P_1) \circledast (L P_2) \neq 0$. □

Remark 4.13 (Partial algebra)*.* $\mathrm{Post}(X)$ is a convex set with a partial, normalized, associative composition and a partial conditioning action. It is not a semiring: $\oplus$ distributes over $\otimes$ only with the reweighting of proposition 4.8.

5 Observational equivalence

5.1 Equivalence and congruence

Definition 5.1. $P_1 \equiv P_2$ iff $\pi_{X*}(P_1) = \pi_{X*}(P_2)$.

Theorem 5.2 (Affine first isomorphism theorem)**.** *π_{X*} is affine. The relation $\equiv$ is a congruence for convex combinations, so $\mathrm{Post}(X)/{\equiv}$ carries a unique convex structure making the quotient map affine, and π_{X*} induces an affine injection $\mathrm{Post}(X)/{\equiv} \hookrightarrow \mathsf{D}(X)$. If $\mathrm{Hyp}(X) \neq \emptyset$ it is a bijection, so $\mathrm{Post}(X)/{\equiv} \cong \mathsf{D}(X)$ as convex sets.*

Proof. $\pi_{X*} = \mathsf{D}(\mathrm{pr}_1)$ is affine. If $P_i \equiv P_i'$ then $\pi_{X*}(\sum a_i P_i) = \sum a_i \pi_{X*}(P_i) = \pi_{X*}(\sum a_i P_i')$, so $\equiv$ is a congruence and the induced map is well defined, affine and injective. For surjectivity pick $h_0 \in \mathrm{Hyp}(X)$; then $\pi_{X*}(\sum_x \rho(x) \delta_{(x, h_0)}) = \rho$. □

Proposition 5.3 (Congruence for $\oplus$)**.** *If $P_1 \equiv P_1'$ and $P_2 \equiv P_2'$ then $P_1 \oplus_\lambda P_2 \equiv P_1' \oplus_\lambda P_2'$.*

Proposition 5.4 (Congruence for $\otimes$)**.** *If $\perp$ is total, then $\pi_{X*}(P_1 \otimes P_2) = \mathcal{N}(m_1 m_2)$ with $m_i = \pi_{X*}(P_i)$, so $\equiv$ is a congruence for $\otimes$ (no condition on q is needed). If $\perp$ is not total this can fail.*

Proof. With $\perp$ total, $\sum_h (P_1 \circledast P_2)(x,h) = \sum_{h_1,h_2} P_1(x,h_1)P_2(x,h_2) = m_1(x)m_2(x)$. For the failure, take $X = \{x,y\}$, hypotheses h_1, h_2, h_3 with $h_1 \perp h_3$ but not $h_2 \perp h_3$, and $P_1 = \frac{1}{2}\delta_{(x,h_1)} + \frac{1}{2}\delta_{(y,h_2)}$, $P_1' = \frac{1}{2}\delta_{(x,h_2)} + \frac{1}{2}\delta_{(y,h_1)}$, $P_2 = \frac{1}{2}\delta_{(x,h_3)} + \frac{1}{2}\delta_{(y,h_3)}$. Then $P_1 \equiv P_1'$ but the marginals of $P_1 \otimes P_2$ and $P_1' \otimes P_2$ are δ_x and δ_y. □

Proposition 5.5 (Congruence for Cond)**.** *Let $|X| \geq 2$. Then $\equiv$ is a congruence for Cond_e (on all P where both sides are defined) if and only if $L(e \mid \cdot)$ is constant on $\mathrm{Hyp}(X)$.*

Proof. If L is constant and positive, $\mathrm{Cond}_e = \mathrm{id}$. Conversely, suppose $L(e \mid h_1) = \ell_1 \neq \ell_2 = L(e \mid h_2)$, so $\ell_1 + \ell_2 > 0$. With distinct $x_1, x_2 \in X$ let $P = \frac{1}{2}\delta_{(x_1,h_1)} + \frac{1}{2}\delta_{(x_2,h_2)}$ and $P' = \frac{1}{2}\delta_{(x_1,h_2)} + \frac{1}{2}\delta_{(x_2,h_1)}$. Then $P \equiv P'$, but the marginals of $\mathrm{Cond}_e P$ and $\mathrm{Cond}_e P'$ are $(\ell_1,\ell_2)/(\ell_1+\ell_2)$ and $(\ell_2,\ell_1)/(\ell_1+\ell_2)$, which differ. □

Remark 5.6. In particular, requiring $L(e \mid h) = \ell(e \mid q_X(h))$ does *not* make $\equiv$ a congruence for conditioning: in the counterexample $q_X(h_1) \neq q_X(h_2)$ is allowed. With $\ell_1 = 2, \ell_2 = 1$ the conditioned marginals are $(\frac{2}{3},\frac{1}{3})$ and $(\frac{1}{3},\frac{2}{3})$.

5.2 Identifiability of decompositions

Theorem 5.2 says that outcome data determine only the image in $\mathsf{D}(X)$; the hypothesis that produced them is quotiented away. We now ask when the weights of known sources can nevertheless be recovered. In this subsection and the next, X is finite and $A_1, \ldots, A_k \in \mathsf{D}(X)$ are fixed *sources*, for instance the predictives $q_X(h_i)$ of hypotheses or the admissible posteriors S_1 of section 10. For $x \in \mathsf{D}(X)$ the set of *decompositions* of x is

$$W(x) = \Big\{ \lambda \in \Delta_k : \ \textstyle\sum_i \lambda_i A_i = x \Big\},$$

a polytope, nonempty if and only if $x \in \mathrm{conv}(A_1, \ldots, A_k)$. The *exclusive region* of source i is $R_i = \{b \in X : A_i(b) > 0, \ A_j(b) = 0 \text{ for } j \neq i\}$.

Proposition 5.7 (Identifiability)**.**

(i) $W(x)$ has at most one point for every x if and only if $A_1, \ldots, A_k$ are affinely independent. In that case $x \mapsto \lambda(x)$ is affine on $\mathrm{conv}(A_1, \ldots, A_k)$.

(ii) For any sources, if $A_i(R_i) > 0$ then $\lambda_i = x(R_i)/A_i(R_i)$ for every $\lambda \in W(x)$.

(iii) At most $|X|$ sources can be affinely independent.

(iv) For each i, $\{\lambda_i : \lambda \in W(x)\}$ is a closed interval whose endpoints are the values of two linear programs over $W(x)$.

(v) Given $\lambda \in W(x)$, an observation b with $x(b) > 0$ came from source i with probability $\lambda_i A_i(b)/x(b)$; as λ ranges over $W(x)$ this ranges over a closed interval with linear-programming endpoints, reducing to a point when $W(x)$ is a point.

Proof. (i) If $\sum_i \lambda_i A_i = \sum_i \lambda'_i A_i$ with $\lambda, \lambda' \in \Delta_k$, then $c = \lambda - \lambda'$ satisfies $\sum_i c_i A_i = 0$ and $\sum_i c_i = 0$, which forces $c = 0$ under affine independence. Conversely, an affine dependence $c \neq 0$ gives, at $x = \sum_i \frac{1}{k} A_i$, the two decompositions $\frac{1}{k}\mathbf{1} \pm tc$ for small $t > 0$. The inverse of an injective affine map is affine. (ii) On R_i only source i has mass, so $x(R_i) = \lambda_i A_i(R_i)$. (iii) Affinely independent points of $\mathsf{D}(X)$, an affine space of dimension $|X| - 1$, number at most $|X|$. (iv) $W(x)$ is a polytope and $\lambda \mapsto \lambda_i$ is linear. (v) $x(b)$ is fixed, so the probability is linear in λ. □

The data of example 10.16(b) give an instance of non-identifiability: for the sources δ_0, δ_1, u on $X = \{0, 1\}$, the set $W(u)$ is the segment joining the weight vectors $(0, 0, 1)$ and $(\frac{1}{2}, \frac{1}{2}, 0)$. Even when $W(x)$ is a point, attribution of a *single* observation in an overlap of supports is genuinely probabilistic, as in (v); what is exact is the attribution of the mixture as a whole, namely the measures $\lambda_i A_i$.

Proposition 5.8 (Supports only)**.** *Suppose only the supports of two sources are known:* $\operatorname{supp} A \subseteq S_A$, $\operatorname{supp} B \subseteq S_B$, *with* $\operatorname{supp} x \subseteq S_A \cup S_B$. *Put* $R_A = S_A \setminus S_B$, $R_B = S_B \setminus S_A$ *and* $R_{AB} = S_A \cap S_B$. *Then the set of* λ *for which there exist* A, B *with these supports and* $x = \lambda A + (1 - \lambda)B$ *is exactly*

$$\big[\, x(R_A),\ 1 - x(R_B) \,\big].$$

This interval is $[\mathrm{Bel}(\{A\}), \mathrm{Pl}(\{A\})]$ *for the Dempster–Shafer mass function* $m(\{A\}) = x(R_A)$, $m(\{B\}) = x(R_B)$, $m(\{A, B\}) = x(R_{AB})$ *[21, 41]. More generally, for* k *sources with known supports, let* $T(b)$ *be the set of sources whose support contains* b *and* $m(T) = x(\{b : T(b) = T\})$*; the feasible weight vectors are exactly the core* $\{\lambda \in \Delta_k : \sum_{i \in T} \lambda_i \geq \mathrm{Bel}(T)$ *for all* $T\}$ *of the belief function.*

Proof. On R_A the source B vanishes, so $\lambda A = x$ there and $\lambda \geq x(R_A)$; symmetrically $1 - \lambda \geq x(R_B)$. Conversely, for λ in the interval put $a = \lambda - x(R_A) \geq 0$ and $b = 1 - \lambda - x(R_B) \geq 0$, so $a + b = x(R_{AB})$; let λA equal x on R_A and $\frac{a}{x(R_{AB})} x$ on R_{AB}, and $(1 - \lambda)B$ equal x on R_B and $\frac{b}{x(R_{AB})} x$ on R_{AB} (both zero on R_{AB} if $x(R_{AB}) = 0$). For k sources, a weight vector is feasible iff the mass $x(b)$ at each outcome can be allocated to the sources in $T(b)$ with totals λ_i. By Gale's supply–demand theorem [27] this holds iff $\sum_{i \in S} \lambda_i \leq x(\{b : T(b) \cap S \neq \emptyset\}) = \mathrm{Pl}(S)$ for every set S of sources, which, since $\sum_i \lambda_i = 1$, is the core condition applied to the complement of S. □

The three regimes are summarized as follows.

Known about the sources	Weights	Attribution of overlap mass
full distributions, affinely independent	unique (proposition 5.7(i))	exact probability
full distributions, affinely dependent	polytope, LP bounds	interval
supports only	core of a belief function	belief–plausibility interval

5.3 Robust decomposition under unmodeled contamination

All three regimes assume that the declared sources are complete. To allow for sources outside the model we use the contamination model of Huber [32]:

$$x = \sum_i \lambda_i A_i + \mu U, \qquad \lambda \geq 0,\ \mu \geq 0,\ \sum_i \lambda_i + \mu = 1, \qquad U \in \mathsf{D}(X) \text{ unknown.}$$

Lemma 5.9. (λ, μ) *is compatible with* x *for some* U *if and only if* $\sum_i \lambda_i A_i \leq x$ *pointwise and* $\sum_i \lambda_i = 1 - \mu$. *Then* $U = (x - \sum_i \lambda_i A_i)/\mu$ *when* $\mu > 0$.

Proof. $\mu U = x - \sum_i \lambda_i A_i$ must be a nonnegative measure of mass μ. If $\mu = 0$, a nonnegative measure of mass zero vanishes. □

Theorem 5.10 (Unmodeled weight)**.** *Let* $\mu_{\min}(x) = 1 - \max\{\sum_i \lambda_i : \lambda \geq 0,\ \sum_i \lambda_i A_i \leq x\}$.

(i) The unmodeled weights compatible with x form the interval $[\mu_{\min}(x), 1]$.

(ii) (Certificate) $1 - \mu_{\min}(x) = \min\{\langle y, x\rangle :\ y \geq 0,\ \langle y, A_i\rangle \geq 1 \text{ for all } i\}$. *Every such y certifies* $\mu \geq 1 - \langle y, x\rangle$.

(iii) $\mu_{\min}(x) = 0$ *if and only if* $x \in \operatorname{conv}(A_1, \ldots, A_k)$*; the level-one test of theorem 10.8 is the case of zero budget.*

(iv) (Budget) Under an assumed budget $\mu \leq \bar{\mu}$*, x is explainable iff* $\mu_{\min}(x) \leq \bar{\mu}$*, and sharp bounds on each λ_i, on μ, and on the unmodeled mass* $x(b) - \sum_i \lambda_i A_i(b)$ *at each outcome are the values of linear programs over* $\{\lambda \geq 0 : \sum_i \lambda_i A_i \leq x,\ \sum_i \lambda_i \geq 1 - \bar{\mu}\}$.

Proof. (i) By lemma 5.9, compatible μ are $1 - \sum_i \lambda_i$ over the feasible λ; if λ is feasible so is $t\lambda$ for $t \in [0, 1]$. (ii) Linear programming duality; the primal is feasible at $\lambda = 0$ and the dual at a large constant y. (iii) If $\sum_i \lambda_i = 1$ and $\sum_i \lambda_i A_i \leq x$, both sides have mass one, so they are equal. (iv) Direct from lemma 5.9. □

Remark 5.11 (Mixture proportion estimation and contaminated sampling). With a single source A, $1 - \mu_{\min}(x) = \min_{b \in \operatorname{supp} A} x(b)/A(b)$ is the maximal proportion of A in x studied in mixture proportion estimation [10, 40], and the budget $\bar{\mu}$ of (iv) plays the role of the known bound on the error probability in the sharp bounds of Horowitz and Manski for contaminated sampling [30]. Those works estimate from samples under identifying assumptions; here x is known and the whole set of decompositions is described, with certificates. The companion paper [3] develops this into a calculus of attribution for model interventions.

Proposition 5.12 (The distance test is one-sided)**.** $d(x, \operatorname{conv}(A_1, \ldots, A_k)) \leq 2\mu_{\min}(x)$ *in ℓ^1, so $d > 2\bar{\mu}$ certifies that x is not explainable within budget $\bar{\mu}$. The converse fails: for one uniform source on three outcomes and* $x = (\frac{1}{2}, \frac{1}{2}, 0)$, $d = \frac{2}{3}$ *while* $\mu_{\min}(x) = 1$*, certified by* $y = (0, 0, 3)$.

Proof. If $x = s + \mu U$ with $s = \sum_i \lambda_i A_i$ of mass $1 - \mu$ and $\mu < 1$, put $a = s/(1 - \mu) \in \operatorname{conv}(A)$; then $x - a = \mu(U - a)$ has norm at most 2μ (if $\mu = 1$ the bound $d \leq 2$ is trivial). In the example, x vanishes where the source has mass $\frac{1}{3}$, so no positive multiple of the source fits under x; and $\langle y, A\rangle = 1$, $\langle y, x\rangle = 0$. □

Proposition 5.13 (Counting is not identifiable)**.** *If unmodeled sources may be arbitrary elements of* $\mathsf{D}(X)$*, then x is compatible with m unmodeled sources of total weight ν if and only if it is compatible with one unmodeled source of weight ν.*

Proof. $\sum_j \nu_j U_j = \nu U$ with $U = \sum_j (\nu_j/\nu) U_j \in \mathsf{D}(X)$, and conversely take $m - 1$ copies of weight zero. □

Remark 5.14 (What is certified). From the data alone one obtains $\mu_{\min}(x)$: at least this fraction of x is not explained by the declared sources, with the explicit certificate y of theorem 5.10(ii). Every upper bound, and every rejection, depends on the assumed budget $\bar{\mu}$; with $\bar{\mu} = 1$ nothing is rejected. A certificate can show that the unmodeled weight exceeds a budget, but by proposition 5.13 not that more than one unmodeled mechanism is present. Counting becomes meaningful only when unmodeled sources are restricted to a family that is not closed under mixing; for point masses,

for instance, the least number of unmodeled sources is the least support size of $x - \sum_i \lambda_i A_i$ over compatible λ. In short, attribution is complete relative to a declared list of sources and a declared budget, and the certificates report when the data contradict them; no test can detect an unmodeled source whose contribution lies inside the declared convex hull.

6 The three-level tower and the context comonad

6.1 The three posterior levels

For a set X let $H_3(X) = \mathsf{D}(X) \times \mathsf{D}^2(X) \times \mathsf{D}^3(X)$, with elements (ρ, Π, Ω). The barycentric projections $b_X = \mu_X \colon \mathsf{D}^2 X \to \mathsf{D}X$ and $b_X^{(2)} = \mu_{\mathsf{D}X} \colon \mathsf{D}^3 X \to \mathsf{D}^2 X$ form the tower $\mathsf{D}^3 X \to \mathsf{D}^2 X \to \mathsf{D}X$. By monad associativity $\mu_X \circ \mu_{\mathsf{D}X} = \mu_X \circ \mathsf{D}(\mu_X)$, so the full collapse $\mathsf{D}^3 X \to \mathsf{D}X$ does not depend on which projection is used at the top. The three levels are distinct state; the projections are observation maps.

6.2 A general context comonad

For a set X, an element c of a set A, and $y \in X$, write $j_{X,c}(y) = (y, c)$.

Theorem 6.1 (Context comonad)**.** *Let* $\mathsf{E} \colon \mathbf{Set} \to \mathbf{Set}$ *be any functor. Put* $\mathsf{C}_\mathsf{E}(X) = X \times \mathsf{E}(X)$ *and* $\mathsf{C}_\mathsf{E}(f) = f \times \mathsf{E}(f)$*, and define*

$$\varepsilon_X(x, c) = x, \qquad \delta_X(x, c) = \big((x, c),\ \kappa_X(c)\big), \qquad \kappa_X(c) = \mathsf{E}(j_{X,c})(c) \in \mathsf{E}(\mathsf{C}_\mathsf{E} X),$$

where $j_{X,c} \colon X \to \mathsf{C}_\mathsf{E} X$*. Then* $(\mathsf{C}_\mathsf{E}, \varepsilon, \delta)$ *is a comonad on* $\mathbf{Set}$*.*

Proof. Write $\mathsf{C} = \mathsf{C}_\mathsf{E}$. Functoriality is that of $\mathrm{id} \times \mathsf{E}$, and ε is natural because it is a product projection.

Naturality of δ. For $f \colon X \to Y$ we need $\delta_Y(fx, \mathsf{E}f\, c) = \mathsf{C}\mathsf{C}f(\delta_X(x, c))$. The second component of the right side is $\mathsf{E}(\mathsf{C}f)(\mathsf{E}(j_{X,c})(c)) = \mathsf{E}(\mathsf{C}f \circ j_{X,c})(c)$. Now $(\mathsf{C}f \circ j_{X,c})(y) = (fy, \mathsf{E}f\, c) = (j_{Y,\mathsf{E}f c} \circ f)(y)$, so this equals $\mathsf{E}(j_{Y,\mathsf{E}f c})(\mathsf{E}f\, c) = \kappa_Y(\mathsf{E}f\, c)$. First components agree trivially.

Counit laws. $\varepsilon_{\mathsf{C}X} \delta_X(x, c) = (x, c)$. Also $\mathsf{C}(\varepsilon_X)\delta_X(x, c) = (\varepsilon_X(x, c), \mathsf{E}(\varepsilon_X)\mathsf{E}(j_{X,c})(c)) = (x, c)$ since $\varepsilon_X \circ j_{X,c} = \mathrm{id}_X$.

Coassociativity. Let $w = (x, c)$ and $c' = \kappa_X(c)$. We first note

$$\delta_X \circ j_{X,c} = j_{\mathsf{C}X,c'} \circ j_{X,c}, \tag{1}$$

since both send y to $((y, c), c')$; this uses that $\kappa_X(c)$ depends only on c. Now $\delta_{\mathsf{C}X}\delta_X(x, c) = \big((w, c'), \mathsf{E}(j_{\mathsf{C}X,c'})(c')\big) = \big((w, c'), \mathsf{E}(j_{\mathsf{C}X,c'} \circ j_{X,c})(c)\big)$, while $\mathsf{C}(\delta_X)\delta_X(x, c) = \big(\delta_X(w), \mathsf{E}(\delta_X)(c')\big) = \big((w, c'), \mathsf{E}(\delta_X \circ j_{X,c})(c)\big)$. These agree by (1). □

Remark 6.2. If E is constant at a set A, then C_E is the reader (product) comonad $X \times A$. For general E the context is transported into $\mathsf{C}_\mathsf{E} X$ by κ_X rather than copied.

6.3 The three-level context

Definition 6.3. $\mathsf{E}_3(X) = \mathsf{D}(X) \times \mathsf{D}^2(X) \times \mathsf{D}^3(X) \times \mathrm{Sh}(X)$ and $\mathsf{C}_3 = \mathsf{C}_{\mathsf{E}_3}$, so $\mathsf{C}_3(X) = X \times \mathsf{D}X \times \mathsf{D}^2 X \times \mathsf{D}^3 X \times \mathrm{Sh}X$ with points $(x, \rho, \Pi, \Omega, \Sigma)$.

Corollary 6.4. C_3 *is a comonad. Explicitly, for* $c = (\rho, \Pi, \Omega, \Sigma)$*,*

$$\kappa_X(c) = \big((j_{X,c})_* \rho,\ \mathsf{D}^2(j_{X,c})(\Pi),\ \mathsf{D}^3(j_{X,c})(\Omega),\ \mathsf{D}(\mathrm{Hyp}(j_{X,c}))(\Sigma)\big).$$

Each component of $\mathsf{E}_3(Y)$ is of the form $\mathsf{D}(Z)$, so $\mathsf{E}_3(Y)$ is a convex set, and for every f the map $\mathsf{E}_3(f)$ is affine.

Remark 6.5 (The absolute tower grows under D). Because the levels of C_3 are indexed by the object, $\mathsf{C}_3(\mathsf{D}X) = \mathsf{D}X \times \mathsf{D}(\mathsf{D}X) \times \mathsf{D}^2(\mathsf{D}X) \times \mathsf{D}^3(\mathsf{D}X) \times \mathrm{Sh}(\mathsf{D}X) = \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{D}^4X \times \mathrm{Sh}(\mathsf{D}X)$. A transformation $\mathsf{D}\mathsf{C}_3X \to \mathsf{C}_3\mathsf{D}X$ must therefore supply a fourth-order component, and applying D again raises the order once more. In the notation of the first version, the components $P_M \in \mathsf{D}X$, $\Pi_M \in \mathsf{D}^2X$ and $\Omega_M \in \mathsf{D}^3X$ fill only the first three factors. The transport in corollary 6.4 is $\rho \mapsto (j_{X,c})_*\rho$, $\Pi \mapsto \mathsf{D}^2(j_{X,c})\Pi$, $\Omega \mapsto \mathsf{D}^3(j_{X,c})\Omega$; it is this re-indexing of the context over C_3X that makes the order grow.

6.4 The context lift

Definition 6.6. $J_{X,c} = \mathrm{Hyp}(j_{X,c})\colon \mathrm{Hyp}(X) \to \mathrm{Hyp}(\mathsf{C}_3X)$; explicitly $J_{X,c}(\theta,\rho) = (\theta, (j_{X,c})_*\rho)$.

Proposition 6.7 (Coherence of J). *(CL1)* $q_{\mathsf{C}_3X} \circ J_{X,c} = (j_{X,c})_* \circ q_X$;

(CL2) $\mathrm{Hyp}(\varepsilon_X) \circ J_{X,c} = \mathrm{id}_{\mathrm{Hyp}(X)}$;

(CL3) $\mathrm{Hyp}(\delta_X) \circ J_{X,c} = J_{\mathsf{C}_3X,\kappa_X(c)} \circ J_{X,c}$.
The shadow component of $\kappa_X(c)$ is $(J_{X,c})_*\Sigma$.

Proof. (CL1) is naturality of q. (CL2) is functoriality of Hyp and $\varepsilon_X \circ j_{X,c} = \mathrm{id}$. (CL3) is functoriality of Hyp applied to (1). □

7 The distribution-level lift

For $k \geq 1$ let $\eta_X^{(k)} = \eta_{\mathsf{D}^kX} \circ \cdots \circ \eta_{\mathsf{D}X}\colon \mathsf{D}X \to \mathsf{D}^{k+1}X$, so $\eta_X^{(3)}(\rho) = \delta_{\delta_{\delta_\rho}}$.

Definition 7.1. $K_X^{[k]}\colon \mathrm{Hyp}(X) \to \mathrm{Hyp}(\mathsf{D}^kX)$, $K_X^{[k]}(\theta,\rho) = (\theta, \eta_X^{(k)}(\rho))$. Then $q_{\mathsf{D}^kX}(K_X^{[k]}h) = \eta_X^{(k)}(q_Xh)$; for $k=3$ this is $\delta_{\delta_{\delta_{\rho_h}}}$.

Proposition 7.2 (Naturality and K/J coherence). *$K^{[k]}\colon \mathrm{Hyp} \Rightarrow \mathrm{Hyp} \circ \mathsf{D}^k$ is natural. In particular*

$$\mathrm{Hyp}(\mathsf{D}^3 j_{X,c}) \circ K_X^{[3]} = K_{\mathsf{C}_3X}^{[3]} \circ J_{X,c}.$$

Proof. $\eta^{(k)}$ is a composite of whiskered units, hence natural: $\mathsf{D}^{k+1}(f) \circ \eta_X^{(k)} = \eta_Y^{(k)} \circ \mathsf{D}(f)$. Apply with $f = j_{X,c}$. □

Remark 7.3. The coherence must use $\mathsf{D}^3 j_{X,c}$: $K_X^{[3]}$ lands in $\mathrm{Hyp}(\mathsf{D}^3X)$, on which $\mathrm{Hyp}(\mathsf{D}^2 j)$ does not act. Note also that $K_X^{[3]}(h)$ is a point mass at the single point $\delta_{\delta_{\rho_h}}$ of D^3X; it is not the pushforward of ρ_h along $x \mapsto \delta_{\delta_{\delta_x}}$, which would be $\mathrm{Hyp}(\eta\eta\eta)$.

8 Option 1: mixed distributive laws for the absolute tower

Throughout, $M = \sum_i a_i \delta_{(x_i,c_i)} \in \mathsf{D}\mathsf{C}_3X$ with $c_i = (\rho_i, \Pi_i, \Omega_i, \Sigma_i)$. The target is $\mathsf{C}_3\mathsf{D}X = \mathsf{D}X \times \mathsf{D}(\mathsf{D}X) \times \mathsf{D}^2(\mathsf{D}X) \times \mathsf{D}^3(\mathsf{D}X) \times \mathrm{Sh}(\mathsf{D}X)$.

8.1 What the unit axiom forces

Lemma 8.1. *If $\lambda\colon \mathsf{DC}_3 \Rightarrow \mathsf{C}_3\mathsf{D}$ satisfies (U), then $\lambda(\delta_{(x,c)}) = (\delta_x, \mathsf{E}_3(\eta_X)(c))$. In particular its level-one component is $\mathsf{D}(\eta_X)(\rho) = \sum_y \rho(y)\,\delta_{\delta_y}$, which differs from δ_ρ whenever ρ is not a point mass. The two agree after barycentric collapse: $\mu_X(\mathsf{D}(\eta_X)\rho) = \rho = \mu_X(\delta_\rho)$.*

Proof. (U) states $\lambda(\eta_{\mathsf{C}_3X}(x,c)) = \mathsf{C}_3(\eta_X)(x,c) = (\eta_X x, \mathsf{E}_3(\eta_X)c)$. □

So a law that keeps a non-degenerate posterior ρ as a single hypothesis δ_ρ at level one cannot satisfy (U); the unit axiom requires level-one information to be spread over point hypotheses. This is the precise form of the tension between hierarchy and barycentric collapse.

8.2 The shift law

Definition 8.2 (Shift law)**.** With $K_X^{[1]}(\theta,\rho) = (\theta, \delta_\rho)$,

$$\lambda^\sigma_X(M) = \Big(\sum_i a_i\delta_{x_i},\ \sum_i a_i\delta_{\rho_i},\ \sum_i a_i\delta_{\Pi_i},\ \sum_i a_i\delta_{\Omega_i},\ \sum_i a_i\,(K_X^{[1]})_*\Sigma_i\Big).$$

The components lie in $\mathsf{D}X$, $\mathsf{D}(\mathsf{D}X)$, $\mathsf{D}(\mathsf{D}^2X) = \mathsf{D}^2(\mathsf{D}X)$, $\mathsf{D}(\mathsf{D}^3X) = \mathsf{D}^3(\mathsf{D}X)$ and $\mathsf{D}(\mathrm{Hyp}\mathsf{D}X) = \mathrm{Sh}(\mathsf{D}X)$ respectively, so $\lambda^\sigma_X\colon \mathsf{DC}_3X \to \mathsf{C}_3\mathsf{D}X$.

Theorem 8.3. *λ^σ is natural and satisfies (Co). It violates (U), (M) and (Cm). The defects in (U) and (M) at level one vanish after applying μ_X.*

Proof. Naturality. Each component of λ^σ is the pushforward of M along a natural projection (followed, for the shadow, by the natural map $K^{[1]}$ and barycenter, both natural).

(Co). $\varepsilon_{\mathsf{D}X}(\lambda^\sigma M) = \sum_i a_i\delta_{x_i} = \mathsf{D}(\varepsilon_X)(M)$.

(U) fails. By lemma 8.1: $\lambda^\sigma(\delta_{(x,c)})$ has level one δ_ρ, while $\mathsf{C}_3(\eta_X)(x,c)$ has $\mathsf{D}(\eta_X)\rho$. For $X = \{0,1\}$ and $\rho = \frac12\delta_0 + \frac12\delta_1$ these are δ_ρ and $\frac12\delta_{\delta_0} + \frac12\delta_{\delta_1}$.

(M) fails. Let $M' = \sum_\ell \nu_\ell\delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k}\delta_{(x_{\ell k}, c_{\ell k})}$. The level-one component of $\lambda^\sigma\mu(M')$ is $\sum_{\ell,k}\nu_\ell a_{\ell k}\delta_{\rho_{\ell k}}$. On the other side, $\lambda^\sigma_{\mathsf{D}X}$ reads the level-one components $\sum_k a_{\ell k}\delta_{\rho_{\ell k}}$ of $\lambda^\sigma(M_\ell)$ and $\mathsf{C}_3(\mu_X)$ applies $\mathsf{D}(\mu_X)$, giving $\sum_\ell \nu_\ell\delta_{\sum_k a_{\ell k}\rho_{\ell k}}$. These differ whenever the $\rho_{\ell k}$ for fixed ℓ are not all equal, and both have barycenter $\sum_{\ell,k}\nu_\ell a_{\ell k}\rho_{\ell k}$.

(Cm) fails. Take $M = \frac12\delta_{(x,c_1)} + \frac12\delta_{(x,c_2)}$ with $\rho_1 \neq \rho_2$. The level-one component of $\delta_{\mathsf{D}X}(\lambda^\sigma M)$ is $\mathsf{D}(j_{\mathsf{D}X,e})(\sum_i \frac12\delta_{\rho_i}) = \sum_i \frac12\delta_{(\rho_i,e)}$, where e is the pooled context of $\lambda^\sigma M$. On the other side, $\mathsf{C}_3(\lambda^\sigma)\circ\lambda^\sigma_{\mathsf{C}_3X}\circ\mathsf{D}(\delta_X)$ gives $\sum_i \frac12\delta_{\lambda^\sigma((j_{X,c_i})_*\rho_i)} = \sum_i \frac12\delta_{(\rho_i,e_i)}$ with $e_i = (\delta_{\rho_i}, \delta_{\Pi_i}, \delta_{\Omega_i}, (K^{[1]})_*\Sigma_i)$, and $e_i \neq e$. □

Remark 8.4. The transformation $\lambda^{[3]}$ of the earlier version is λ^σ with the level-three component omitted and a shadow component of the wrong type. Its claimed unit and comultiplication theorems fail by theorem 8.3.

8.3 The averaging law

Let $\beta\colon \mathsf{D}(\mathsf{E}_3Y) \to \mathsf{E}_3Y$ be the barycenter, computed componentwise by μ on each factor $\mathsf{D}(Z)$; every $\mathsf{E}_3(f)$ commutes with β because it is affine.

Definition 8.5 (Averaging law)**.**

$$\bar\lambda_X(M) = \Big(\sum_i a_i\delta_{x_i},\ \sum_i a_i\,\mathsf{E}_3(\eta_X)(c_i)\Big),$$

i.e. $\bar\lambda_X = \big\langle \mathsf{D}(\varepsilon_X),\ \beta\circ\mathsf{D}(\mathsf{E}_3(\eta_X)\circ\mathrm{pr}_2)\big\rangle$.

	natural	(U)	(Co)	(M)	(Cm)	keeps δ_ρ at level one
shift law λ^σ	yes	no	yes	no	no	yes
averaging law $\bar\lambda$	yes	yes	yes	yes	no	no

Table 1: Status of the two mixed distributive laws $\mathsf{DC}_3 \Rightarrow \mathsf{C}_3\mathsf{D}$ (theorems 8.3, 8.6 and 8.7). By lemma 8.1, no law satisfying (U) keeps δ_ρ at level one.

Theorem 8.6. *$\bar\lambda$ is natural and satisfies (U), (Co) and (M) exactly.*

Proof. Naturality. For $f\colon X \to Y$, using that $\mathsf{E}_3(\mathsf{D}f)$ is affine and $\mathsf{D}f \circ \eta_X = \eta_Y \circ f$,

$$\mathsf{E}_3(\mathsf{D}f)\Big(\sum_i a_i \mathsf{E}_3(\eta_X)c_i\Big) = \sum_i a_i \mathsf{E}_3(\eta_Y)(\mathsf{E}_3(f)c_i),$$

which is the context component of $\bar\lambda_Y(\mathsf{D}(\mathsf{C}_3 f)M)$; first components agree by naturality of $\mathsf{D}(\varepsilon)$.

(U). $\bar\lambda(\delta_{(x,c)}) = (\delta_x, \mathsf{E}_3(\eta_X)c) = \mathsf{C}_3(\eta_X)(x,c)$.

(Co). $\varepsilon_{\mathsf{D}X} \circ \bar\lambda_X = \mathsf{D}(\varepsilon_X)$ by definition.

(M). Let $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k}\delta_{(x_{\ell k}, c_{\ell k})}$, and put

$$P_\ell = \sum_k a_{\ell k}\delta_{x_{\ell k}}, \qquad e_\ell = \sum_k a_{\ell k}\,\mathsf{E}_3(\eta_X)c_{\ell k}.$$

The left side is $\big(\sum_\ell \nu_\ell P_\ell,\ \sum_\ell \nu_\ell e_\ell\big)$. On the right, $\mathsf{D}(\bar\lambda)M' = \sum_\ell \nu_\ell \delta_{(P_\ell, e_\ell)}$; then $\bar\lambda_{\mathsf{D}X}$ gives

$$\Big(\sum_\ell \nu_\ell \delta_{P_\ell},\ \sum_\ell \nu_\ell\,\mathsf{E}_3(\eta_{\mathsf{D}X})e_\ell\Big),$$

and $\mathsf{C}_3(\mu_X)$ turns this into

$$\Big(\sum_\ell \nu_\ell P_\ell,\ \sum_\ell \nu_\ell\,\mathsf{E}_3(\mu_X \circ \eta_{\mathsf{D}X})e_\ell\Big) = \Big(\sum_\ell \nu_\ell P_\ell,\ \sum_\ell \nu_\ell e_\ell\Big),$$

using that $\mathsf{E}_3(\mu_X)$ is affine and $\mu_X \circ \eta_{\mathsf{D}X} = \mathrm{id}$. □

Theorem 8.7. *$\bar\lambda$ violates (Cm).*

Proof. Let $X = \{0,1\}$, fix Π, Ω, Σ, and let $c_i = (\delta_i, \Pi, \Omega, \Sigma)$ for $i = 0, 1$, $M = \frac{1}{2}\delta_{(0,c_0)} + \frac{1}{2}\delta_{(0,c_1)}$. Write $e_i = \mathsf{E}_3(\eta_X)c_i$ and $e = \frac{1}{2}e_0 + \frac{1}{2}e_1$, so $\bar\lambda M = (\delta_0, e)$ and the level-one component of e is $\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$. The level-one component of $\delta_{\mathsf{D}X}(\bar\lambda M)$ is $\mathsf{D}(j_{\mathsf{D}X,e})$ of that, namely $\frac{1}{2}\delta_{(\delta_0,e)} + \frac{1}{2}\delta_{(\delta_1,e)}$.

On the other side, $\mathsf{D}(\delta_X)M = \sum_i \frac{1}{2}\delta_{((0,c_i),\kappa_X(c_i))}$. Since $\mathsf{E}_3(\bar\lambda_X)$ is affine and $\bar\lambda_X \circ \eta_{\mathsf{C}_3 X} \circ j_{X,c_i} = j_{\mathsf{D}X,e_i} \circ \eta_X$ (by (U)), the context component of $\mathsf{C}_3(\bar\lambda)\bar\lambda_{\mathsf{C}_3 X}\mathsf{D}(\delta_X)(M)$ is $\sum_i \frac{1}{2}\mathsf{E}_3(j_{\mathsf{D}X,e_i})(e_i)$, whose level-one component is $\frac{1}{2}\delta_{(\delta_0,e_0)} + \frac{1}{2}\delta_{(\delta_1,e_1)}$. Since $e_0 \neq e \neq e_1$, the two sides differ. □

Remark 8.8. In general the context component of the right side of (Cm) is $\sum_i a_i \mathsf{E}_3(j_{\mathsf{D}X,e_i})(e_i)$ and that of the left side is $\mathsf{E}_3(j_{\mathsf{D}X,e})(e)$ with $e = \sum_i a_i e_i$: comultiplication copies a *single* pooled context into $\mathsf{C}_3\mathsf{D}X$, whereas the right side keeps one context per sample. Both sides do agree after $\mathsf{C}_3(\varepsilon_{\mathsf{D}X})$ and after $\varepsilon_{\mathsf{C}_3\mathsf{D}X}$, but this follows from (Co) and naturality alone and carries no extra information.

Remark 8.9 (Relation to weak distributive laws). The weak distributive laws of Street and Böhm [44, 16, 17], used by Garner [28] and Goy–Petrişan [29], are laws between two monads that keep the multiplication axioms and weaken the unit axioms; the composite is obtained by splitting an idempotent. Neither λ^σ nor $\bar{\lambda}$ falls under that notion as stated: λ^σ fails a multiplication-type axiom (M), and $\bar{\lambda}$ fails the comonad analogue (Cm). A formulation in which a failing axiom is witnessed by a 2-cell would need a 2-categorical structure on the base (for instance an order or convex enrichment of the relevant hom-sets); on **Set** itself the only 2-cells between natural transformations are identities, so no non-identity 2-cell α is available. The earlier version's claim of a weak distributive law with a barycentric 2-cell is therefore withdrawn. An Section 13.6 shows that the natural candidate order, the convex order, does not supply such 2-cells.

9 Option 2: the base-relative three-level hierarchy

By remark 6.5, the absolute tower cannot hold exactly three levels once the monad is applied. We now fix an outcome base B (in the application, the finite outcome space of CausalIDView) and let the hierarchy live over B regardless of the object at which the comonad is evaluated.

9.1 Coherent contexts

Definition 9.1 (Context algebra)**.** $A_B = \mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B \times \mathrm{Sh}(B)$, with points $c = (\rho, \Pi, \Omega, \Sigma)$. It is a convex set componentwise, and the componentwise barycenter $\beta_B \colon \mathsf{D}(A_B) \to A_B$ (given by μ on each factor $\mathsf{D}Z$) makes (A_B, β_B) an algebra for D, namely the product of the free algebras $(\mathsf{D}Z, \mu_Z)$.

Definition 9.2 (Coherence)**.** A context $c = (\rho, \Pi, \Omega, \Sigma) \in A_B$ is *coherent* if

$$\mu_B(\Pi) = \rho, \qquad \mu_{\mathsf{D}B}(\Omega) = \Pi, \qquad \mathsf{D}(q_B)(\Sigma) = \Pi.$$

Write $A_B^{\mathrm{coh}} \subseteq A_B$ for the coherent contexts.

Coherence says that the barycentric observation maps of the tower $\mathsf{D}^3B \to \mathsf{D}^2B \to \mathsf{D}B$ and the predictive of the shadow agree. The top level determines the lower two, but all three remain state.

Lemma 9.3. *A_B^{coh} is closed under β_B; hence it is a sub-algebra of (A_B, β_B).*

Proof. Each condition is an equation between affine maps (μ_B, $\mu_{\mathsf{D}B}$, $\mathsf{D}(q_B)$ and the projections), and the solution set of such an equation is closed under convex combination. □

9.2 The canonical hierarchy

Definition 9.4. $\sigma_B \colon \mathrm{Sh}(B) \to A_B$,

$$\sigma_B(\Sigma) = \big(\mu_B(\mathsf{D}q_B\,\Sigma),\ \mathsf{D}q_B\,\Sigma,\ \mathsf{D}(\eta_{\mathsf{D}B})(\mathsf{D}q_B\,\Sigma),\ \Sigma\big).$$

On a single hypothesis, $\sigma_B(\delta_h) = (\rho_h,\ \delta_{\rho_h},\ \delta_{\delta_{\rho_h}},\ \delta_h)$, i.e. $\Pi_h = \delta_{\rho_h}$ and $\Omega_h = \delta_{\Pi_h}$.

Proposition 9.5. *σ_B is affine, injective, lands in A_B^{coh}, and is natural in B: $A_f \circ \sigma_B = \sigma_{B'} \circ \mathrm{Sh}(f)$ for $f \colon B \to B'$, where $A_f = \mathsf{D}f \times \mathsf{D}^2f \times \mathsf{D}^3f \times \mathrm{Sh}f$.*

Proof. Each component is a composite of affine maps, and the last component is Σ, so σ_B is injective. Coherence: the first condition holds by definition, the second is the unit law $\mu_{\mathsf{D}B} \circ \mathsf{D}(\eta_{\mathsf{D}B}) = \mathrm{id}$, and the third holds by definition. Naturality follows from the naturality of q, μ and η. □

9.3 The base-relative comonad

Definition 9.6. For a set Y put $\mathsf{C}^B(Y) = Y \times A_B$, $\mathsf{C}^B(f) = f \times \mathrm{id}$, $\varepsilon_Y(y,c) = y$ and $\delta_Y(y,c) = ((y,c),c)$. Write $\mathsf{C}^B_{\mathrm{coh}}$ for the same construction with A_B^{coh}.

This is the product (reader) comonad with parameter A_B, so the comonad laws are immediate. For $Y = B$ the underlying set $\mathsf{C}^B(B) = B \times \mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B \times \mathrm{Sh}(B)$ coincides with $\mathsf{C}_3(B)$. The difference appears under the monad:

$$\mathsf{C}^B(\mathsf{D}Y) = \mathsf{D}Y \times \underbrace{\mathsf{D}B \times \mathsf{D}^2B \times \mathsf{D}^3B}_{\text{still three levels}} \times \mathrm{Sh}(B).$$

9.4 The strict mixed distributive law

Definition 9.7. For $M = \sum_i a_i \delta_{(y_i,c_i)} \in \mathsf{D}\mathsf{C}^B(Y)$,

$$\lambda^B_Y(M) = \Big(\sum_i a_i \delta_{y_i},\ \sum_i a_i c_i \Big) = \big(\mathsf{D}(\mathrm{pr}_1)M,\ \beta_B(\mathsf{D}(\mathrm{pr}_2)M) \big) \in \mathsf{C}^B(\mathsf{D}Y),$$

where $\sum_i a_i c_i$ is the convex combination in A_B, taken level by level:

$$\textstyle\sum_i a_i c_i = \big(\sum_i a_i \rho_i,\ \sum_i a_i \Pi_i,\ \sum_i a_i \Omega_i,\ \sum_i a_i \Sigma_i \big).$$

Theorem 9.8. *$\lambda^B \colon \mathsf{D}\mathsf{C}^B \Rightarrow \mathsf{C}^B\mathsf{D}$ is natural and satisfies all four axioms (U), (M), (Co), (Cm) of definition 2.1. It restricts to a mixed distributive law $\mathsf{D}\mathsf{C}^B_{\mathrm{coh}} \Rightarrow \mathsf{C}^B_{\mathrm{coh}}\mathsf{D}$.*

Proof. Naturality. For $f \colon Y \to Y'$, $\mathsf{D}f \circ \mathsf{D}(\mathrm{pr}_1) = \mathsf{D}(\mathrm{pr}_1) \circ \mathsf{D}(f \times \mathrm{id})$ and the context component is unchanged by $f \times \mathrm{id}$.

(U). $\lambda^B(\delta_{(y,c)}) = (\delta_y, \beta_B(\delta_c)) = (\delta_y, c) = \mathsf{C}^B(\eta_Y)(y,c)$, by the unit law of the algebra β_B.

(Co). $\varepsilon_{\mathsf{D}Y} \circ \lambda^B_Y = \mathsf{D}(\mathrm{pr}_1) = \mathsf{D}(\varepsilon_Y)$.

(M). First components: $\mathsf{D}(\mathrm{pr}_1) \circ \mu_{\mathsf{C}^B Y} = \mu_Y \circ \mathsf{D}^2(\mathrm{pr}_1)$ by naturality of μ, which is the first component of $\mathsf{C}^B(\mu_Y)\lambda^B_{\mathsf{D}Y}\mathsf{D}(\lambda^B_Y)$. Context components: the left side gives $\beta_B \circ \mathsf{D}(\mathrm{pr}_2) \circ \mu = \beta_B \circ \mu_{A_B} \circ \mathsf{D}^2(\mathrm{pr}_2)$; the right side gives $\beta_B \circ \mathsf{D}(\beta_B \circ \mathsf{D}\mathrm{pr}_2) = \beta_B \circ \mathsf{D}(\beta_B) \circ \mathsf{D}^2(\mathrm{pr}_2)$, since $\mathsf{C}^B(\mu_Y)$ does not change the context. These agree by the associativity law $\beta_B \circ \mu_{A_B} = \beta_B \circ \mathsf{D}(\beta_B)$ of the algebra.

(Cm). Let $\bar{c} = \beta_B(\mathsf{D}(\mathrm{pr}_2)M)$. The left side is $\delta_{\mathsf{D}Y}(\lambda^B M) = (\lambda^B M, \bar{c})$. On the right, $\mathsf{D}(\delta_Y)M = \sum_i a_i \delta_{((y_i,c_i),c_i)}$; applying $\lambda^B_{\mathsf{C}^B Y}$ gives $(M, \bar{c})$, and applying $\mathsf{C}^B(\lambda^B_Y)$ gives $(\lambda^B M, \bar{c})$.

Restriction. β_B preserves A_B^{coh} by lemma 9.3. □

Remark 9.9. Theorem 9.8 is an instance of the standard fact that a product comonad $- \times A$ distributes over a monad T whenever A carries a T-algebra structure. The content specific to this paper is the choice of A_B, the coherence sub-algebra and the canonical section σ_B. By the correspondence between mixed distributive laws and liftings [38], D lifts to the category of C^B-coalgebras.

9.5 Pooling, hierarchy and conditioning

Proposition 9.10 (Pooling commutes with hierarchy formation)**.** *If $M = \sum_i a_i \delta_{(y_i, \sigma_B(\Sigma_i))}$, then $\lambda^B(M) = \big(\sum_i a_i \delta_{y_i},\ \sigma_B(\sum_i a_i \Sigma_i) \big)$.*

Proof. σ_B is affine (proposition 9.5). □

Example 9.11 (Distinct posteriors survive at level two)**.** Let $B = \{0, 1\}$, $\rho_0 = \delta_0$, $\rho_1 = \frac{1}{3}\delta_0 + \frac{2}{3}\delta_1$, and let $M = \frac{1}{2}\delta_{(y_0,\sigma(\delta_{h_0}))} + \frac{1}{2}\delta_{(y_1,\sigma(\delta_{h_1}))}$ with $q(h_k) = \rho_k$. The context of $\lambda^B(M)$ has level one $\frac{1}{2}\rho_0 + \frac{1}{2}\rho_1 = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1$ and level two $\frac{1}{2}\delta_{\rho_0} + \frac{1}{2}\delta_{\rho_1}$: the two posteriors are kept as distinct points of $\mathsf{D}B$, and level one is their barycenter, as coherence requires. In contrast, the averaging law of the absolute tower records level one through $\mathsf{D}(\eta)$ (lemma 8.1).

For evidence e with likelihood $L(e \mid \cdot)$ on $\mathrm{Hyp}(B)$, conditioning a shadow is $\mathrm{Cond}_e(\Sigma) = \mathcal{N}(L(e \mid \cdot)\,\Sigma)$ with evidence $Z_e(\Sigma) = \sum_h L(e \mid h)\Sigma(h)$.

Proposition 9.12 (Conditioning commutes with pooling)**.** *Let $M = \sum_i a_i \delta_{(y_i,\sigma_B(\Sigma_i))}$ with $Z_i = Z_e(\Sigma_i) > 0$, and let $Z = \sum_i a_i Z_i$. Define the evidence-reweighted sample*

$$\mathrm{Cond}_e(M) = \sum_i \frac{a_i Z_i}{Z}\, \delta_{(y_i,\sigma_B(\mathrm{Cond}_e\Sigma_i))}.$$

Then the context of $\lambda^B(\mathrm{Cond}_e M)$ is $\sigma_B\big(\mathrm{Cond}_e(\sum_i a_i\Sigma_i)\big)$.

Proof. $L \cdot \sum_i a_i\Sigma_i = \sum_i a_i Z_i \mathrm{Cond}_e(\Sigma_i)$, so $\mathrm{Cond}_e(\sum_i a_i\Sigma_i) = \sum_i (a_i Z_i/Z)\mathrm{Cond}_e(\Sigma_i)$. Apply proposition 9.10. □

9.6 Change of base

Proposition 9.13. *For $f\colon B \to B'$ the map A_f is a morphism of D-algebras that preserves coherence and commutes with σ. The maps $\mathrm{id}_Y \times A_f$ form a comonad morphism $\mathsf{C}^B \Rightarrow \mathsf{C}^{B'}$, and*

$$(\mathrm{id}_{\mathsf{D}Y} \times A_f) \circ \lambda^B_Y = \lambda^{B'}_Y \circ \mathsf{D}(\mathrm{id}_Y \times A_f),$$

so λ^B is natural in the base.

Proof. A_f is a product of maps $\mathsf{D}(g)$, hence affine, hence commutes with β. Coherence is preserved by naturality of μ and q; compatibility with σ is proposition 9.5. The comonad-morphism equations hold because δ copies the context and ε discards it. The displayed equation follows from $A_f \circ \beta_B = \beta_{B'} \circ \mathsf{D}(A_f)$. □

9.7 Stochastic change of base

Proposition 9.13 transports the tower along a function $f\colon B \to B'$. The same holds for a Markov kernel $k\colon B \to \mathsf{D}(B')$, a morphism of the Kleisli category of D. Write $k^{\#}\colon \mathsf{D}B \to \mathsf{D}B'$, $k^{\#}(\rho)(b') = \sum_b \rho(b)\, k(b)(b')$, for its Kleisli extension. Because $\mathrm{Hyp} = \Theta \times \mathsf{D}(-)$, the hypothesis functor extends to kernels by

$$\mathrm{Hyp}(k)(\theta, \rho) = (\theta, k^{\#}\rho),$$

and q stays natural: $q_{B'} \circ \mathrm{Hyp}(k) = k^{\#} \circ q_B$. Put $\mathrm{Sh}(k) = \mathsf{D}(\mathrm{Hyp}(k))$ and

$$A_k = k^{\#} \times \mathsf{D}(k^{\#}) \times \mathsf{D}^2(k^{\#}) \times \mathrm{Sh}(k)\colon A_B \to A_{B'}.$$

For a deterministic $k = \eta_{B'} \circ f$ one has $k^{\#} = \mathsf{D}f$, so $A_k = A_f$.

Proposition 9.14 (Stochastic change of base)**.** *Let $k\colon B \to \mathsf{D}(B')$ and $h\colon B' \to \mathsf{D}(B'')$ be kernels.*

(i) A_k is a morphism of D-algebras $(A_B, \beta_B) \to (A_{B'}, \beta_{B'})$.

(ii) A_k maps A^{coh}_B into $A^{\mathrm{coh}}_{B'}$, and $A_k \circ \sigma_B = \sigma_{B'} \circ \mathrm{Sh}(k)$.

(iii) $A_{h\circ k} = A_h \circ A_k$ *and* $A_{\eta_B} = \mathrm{id}$*, where* $h \circ k$ *is Kleisli composition.*

(iv) The maps $\mathrm{id}_Y \times A_k$ *form a comonad morphism* $\mathsf{C}^B \Rightarrow \mathsf{C}^{B'}$*, and* $(\mathrm{id}_{\mathsf{D}Y} \times A_k) \circ \lambda^B_Y = \lambda^{B'}_Y \circ \mathsf{D}(\mathrm{id}_Y \times A_k)$*.*

Hence $B \mapsto (\mathsf{C}^B, \lambda^B)$ *is functorial on the Kleisli category of* D *restricted to finite sets, extending proposition 9.13.*

Proof. (i) The first component is the Kleisli extension of k, which satisfies $k^\# \circ \mu_B = \mu_{B'} \circ \mathsf{D}(k^\#)$ (it is a morphism of free D-algebras). The other components are pushforwards $\mathsf{D}(g)$ along the functions $g = k^\#$, $\mathsf{D}(k^\#)$ and $\mathrm{Hyp}(k)$, and these commute with μ by naturality. (ii) For a coherent $(\rho, \Pi, \Omega, \Sigma)$: $\mu_{B'}(\mathsf{D}(k^\#)\Pi) = k^\#(\mu_B \Pi) = k^\# \rho$ by (i); $\mu_{\mathsf{D}B'}(\mathsf{D}^2(k^\#)\Omega) = \mathsf{D}(k^\#)(\mu_{\mathsf{D}B}\Omega) = \mathsf{D}(k^\#)\Pi$ by naturality of μ; and $\mathsf{D}(q_{B'})\mathsf{D}(\mathrm{Hyp}k)\Sigma = \mathsf{D}(q_{B'} \circ \mathrm{Hyp}k)\Sigma = \mathsf{D}(k^\# \circ q_B)\Sigma = \mathsf{D}(k^\#)\Pi$. So lemma 9.3 transports. For σ, the first two components follow from the same identities, and the third from naturality of η applied to the function $k^\#$: $\mathsf{D}^2(k^\#) \circ \mathsf{D}(\eta_{\mathsf{D}B}) = \mathsf{D}(\mathsf{D}(k^\#) \circ \eta_{\mathsf{D}B}) = \mathsf{D}(\eta_{\mathsf{D}B'} \circ k^\#)$. So proposition 9.5 transports. (iii) $(h \circ k)^\# = h^\# k^\#$ and $\mathrm{Hyp}(h \circ k) = \mathrm{Hyp}(h)\mathrm{Hyp}(k)$; apply D and D^2. (iv) As in proposition 9.13: δ copies and ε discards the context, and the law commutes with $\mathrm{id} \times A_k$ because $A_k \circ \beta_B = \beta_{B'} \circ \mathsf{D}(A_k)$ by (i). □

Corollary 9.15 (Admissibility and obstructions under stochastic base change)**.** *Transport general admissibility data* (S_1, S_2) *on* B *to* $S'_1 = k^\#(S_1)$ *and* $S'_2 = \mathsf{D}(k^\#)(S_2)$ *on* B'*. Then:*

(i) saturation is preserved, and $L'_1 = k^\#(L_1)$*,* $L'_2 = \mathsf{D}(k^\#)(L_2)$*,* $L'_3 = \mathsf{D}^2(k^\#)(L_3)$*, with the same relations for the effective sets* E_k*;*

(ii) a coherent lift of a claim x *maps to a coherent lift of the transported claim* $k_* x$*; hence* $\varphi(k_* x) \geq \varphi(x)$*, and the inequality can be strict;*

(iii) certificates pull back: if $f' \colon B' \to \mathbb{R}$ *separates* $k^\# x_1$ *from* E'_1*, then* $(k \triangleright f')(b) = \sum_{b'} k(b)(b') f'(b')$ *separates* x_1 *from* E_1*; a level-two certificate* g' *pulls back to* $g' \circ k^\#$*;*

(iv) the secondary obstruction does not increase: $o'_3(\mathsf{D}(k^\#)\Pi) \leq o_3(\Pi)$*.*

Proof. (i) The maps $k^\#$, $\mathsf{D}(k^\#)$ and $\mathsf{D}^2(k^\#)$ are affine and commute with μ (proposition 9.14(i)), the image of a convex hull is the convex hull of the image, and $\mathsf{D}(g)(\mathsf{D}(S)) = \mathsf{D}(g(S))$ for any function g. Saturation: $\mathsf{D}(k^\#)\delta_\rho = \delta_{k^\# \rho}$. (ii) Apply the three maps to the lift and use (i) and proposition 9.14(ii). A lift of $k_* x$ need not come from a lift of x: with $S_1 = \{\delta_0\}$, the claim $x_1 = \delta_1$ fails, but a constant kernel sends both points to the same distribution. (iii) $\langle k \triangleright f', \rho \rangle = \langle f', k^\# \rho \rangle$ for every ρ, so the strict inequality defining a certificate is unchanged. (iv) Pushforward along a function does not increase ℓ^1 distance, and it maps E_2 into E'_2. □

So stochastic base change transports all structure, never manufactures an obstruction, and can destroy one; certificates found downstream are certificates upstream. This is the tower-level form of the fact that precision can only be lost along a 1-cell [3, Prop. 7.3]. The kernels k are the 1-cells of the Kleisli category, and couplings of two kernels are its 2-cells. In [4, Thm. 2.3, Section 3] the context functor $B \mapsto A^{\mathrm{coh}}_B$ is shown to be a discrete opfibration over this category, and the distance between two kernels is shown to bound the distance between transported contexts at every level. A worked example is in [6, Section 11], and the quantitative form for label noise is proposition 11.2.

Proposition 9.16 (Conditioning through a channel)**.** *For an outcome* $e' \in B'$ *observed after the channel* k*, let* $k \triangleright 1_{e'}$ *be the predicate* $h \mapsto (k^\# q_B(h))(e')$ *on* $\mathrm{Hyp}(B)$*. Then* $\mathrm{Cond}_{e'} \circ \mathrm{Sh}(k) = \mathrm{Sh}(k) \circ \mathrm{Cond}_{k \triangleright 1_{e'}}$ *on shadows with positive evidence.*

Proof. Conditioning reweights hypotheses without changing them, and the weight of $\mathrm{Hyp}(k)(h)$ for e' is $q_{B'}(\mathrm{Hyp}(k)h)(e') = (k^{\#}q_B(h))(e')$, which is the weight of h under the pulled-back predicate. Pushforward along a function commutes with a reweighting that factors through it. □

Remark 9.17 (What is not transported). Two things stay as in the deterministic case. The admissible hypotheses $\mathrm{Hyp}^{\mathrm{adm}}$ are not closed under $\mathrm{Hyp}(k)$ (remark 3.4), so admissibility data must be transported as in corollary 9.15 rather than recomputed from descriptors. And conditioning transports only for evidence pulled back along k, as in proposition 9.16; evidence on B has no image on B'.

9.8 Comparison of the two options

At $Y = B$ the two towers have the same points. They differ in the comultiplication: the absolute comonad transports the context into distributions over C_3B through $\rho \mapsto (j_{B,c})_*\rho$, $\Pi \mapsto \mathsf{D}^2(j_{B,c})\Pi$, $\Omega \mapsto \mathsf{D}^3(j_{B,c})\Omega$, whereas the base-relative comonad copies it. That transport is what re-indexes the hierarchy and raises its order under D; dropping it is what allows a strict law.

	levels at $\mathsf{D}Y$	(U)	(Co)	(M)	(Cm)
absolute, shift law λ^{σ}	four	no	yes	no	no
absolute, averaging law $\bar{\lambda}$	four	yes	yes	yes	no
base-relative, λ^{B}	three	yes	yes	yes	yes

Table 2: The two options. Only the base-relative tower keeps exactly three levels and admits a strict mixed distributive law (theorem 9.8).

10 Coherent lifting and obstruction certificates

The base-relative tower of section 9 supports a precise notion of when a claimed posterior state is compatible with a hypothesis class at every level, and of what witnesses a failure. We fix a finite base B throughout.

10.1 Admissibility data and the admissible tower

Definition 10.1 (Admissibility data)**.** Admissibility data consist of a finite set $S_1 \subseteq \mathsf{D}B$ of *admissible posteriors* and a set $S_2 \subseteq \mathsf{D}(S_1) \subseteq \mathsf{D}^2B$ of *admissible second-order states* such that $\delta_\rho \in S_2$ for every $\rho \in S_1$. The *admissible tower* is

$$L_1 = \mathrm{conv}(S_1) \subseteq \mathsf{D}B, \qquad L_2 = \mathsf{D}(S_1) \subseteq \mathsf{D}^2B, \qquad L_3 = \mathsf{D}(S_2) \subseteq \mathsf{D}^3B.$$

Here $\mathsf{D}(S)$ denotes the distributions supported in S.

The basic example comes from a finite admissible hypothesis class $H \subseteq \mathrm{Hyp}(B)$, for instance the hypotheses consistent with the evidence: $S_1 = q_B(H)$. Two choices of S_2 bracket the possibilities. In the *committed regime* $S_2 = \{\delta_\rho : \rho \in S_1\}$: the only admissible second-order states are single admissible posteriors held with certainty. In the *inherited regime* $S_2 = \mathsf{D}(S_1)$: every distribution over admissible posteriors is an admissible state. Richer hypothesis structure, for example composite hypotheses $h_1 \otimes h_2$ with an uncertain component, supplies intermediate choices.

Lemma 10.2. *Each L_k is convex, $\mu_B(L_2) = L_1$ and $\mu_{\mathsf{D}B}(L_3) = L_2$. Moreover* $\mathrm{conv}(S_2) = L_2$.

Proof. Convexity is clear. $\mu_B(\sum_\rho \Pi(\rho)\delta_\rho) = \sum_\rho \Pi(\rho)\rho$, so $\mu_B(\mathsf{D}(S_1)) = \mathrm{conv}(S_1)$. Since $S_2 \subseteq \mathsf{D}(S_1)$ and $\mathsf{D}(S_1)$ is convex, $\mathrm{conv}(S_2) \subseteq \mathsf{D}(S_1)$; conversely every $\Pi \in \mathsf{D}(S_1)$ is $\sum_\rho \Pi(\rho)\delta_\rho$ with each $\delta_\rho \in S_2$. Finally $\mu_{\mathsf{D}B}(\mathsf{D}(S_2)) = \mathrm{conv}(S_2) = L_2$. □

Proposition 10.3 (Canonical contexts realize the committed tower)**.** *Let $S_1 = q_B(H)$ and $S_2 = \{\delta_\rho : \rho \in S_1\}$. As Σ ranges over $\mathsf{D}(H)$, the level components (ρ, Π, Ω) of the canonical contexts $\sigma_B(\Sigma)$ range exactly over the triples $(\mu_B\Pi,\ \Pi,\ \mathsf{D}(\eta_{\mathsf{D}B})\Pi)$ with $\Pi \in L_2$; in particular they fill L_1, L_2 and L_3.*

Proof. $\sigma_B(\Sigma)$ has $\Pi = \mathsf{D}(q_B)\Sigma$, which ranges over $\mathsf{D}(q_B(H)) = L_2$, and $\rho = \mu_B\Pi$, $\Omega = \mathsf{D}(\eta_{\mathsf{D}B})\Pi$. The last map sends $\mathsf{D}(S_1)$ onto $\mathsf{D}(\{\delta_\rho\}) = L_3$. □

10.2 Claims, lifts and failure levels

Definition 10.4 (Claims and lifts)**.** A *claim* is a partial tower $x = (x_1, x_2, x_3)$ with $x_k \in \mathsf{D}^k B \cup \{*\}$ ($*$ meaning unspecified), consistent in the sense that $\mu_B(x_2) = x_1$ and $\mu_{\mathsf{D}B}(x_3) = x_2$ whenever both sides are specified. A *coherent lift* of x is a triple $(\rho, \Pi, \Omega) \in L_1 \times L_2 \times L_3$ with $\mu_B\Pi = \rho$ and $\mu_{\mathsf{D}B}\Omega = \Pi$ that agrees with x on every specified level. The *failure level* of x is

$$\varphi(x) = \min\{k : x_k \neq *,\ x_k \notin L_k\} \in \{1, 2, 3, \infty\}.$$

Theorem 10.5 (Lifting criterion)**.** *Let x be a claim whose highest specified level is m.*

(i) (Monotonicity) If $x_k \notin L_k$ and x_j is specified for some $j > k$, then $x_j \notin L_j$. Hence a claim that fails at level k fails at every specified level above k.

(ii) (Criterion) x has a coherent lift if and only if $x_m \in L_m$, if and only if $\varphi(x) = \infty$.

Proof. (i) By lemma 10.2, $\mu(L_j) \subseteq L_{j-1}$, so if $x_j \in L_j$ then every specified x_i with $i < j$, being an iterated barycenter of x_j, lies in L_i. Contrapose.

(ii) If x has a lift, $x_m \in L_m$. Conversely suppose $x_m \in L_m$. Below m define the lift by barycenters: $\Pi = \mu_{\mathsf{D}B}(x_3)$ if $m = 3$, and $\rho = \mu_B(\Pi)$; these lie in L_2, L_1 by lemma 10.2 and agree with any specified lower x_i by consistency. Above m use explicit sections: if $m \leq 2$ take $\Omega = \mathsf{D}(\eta_{\mathsf{D}B})(\Pi)$, which lies in $\mathsf{D}(\{\delta_\rho : \rho \in S_1\}) \subseteq L_3$ and satisfies $\mu_{\mathsf{D}B}\Omega = \Pi$; if $m = 1$ first write $x_1 = \sum_\rho w_\rho \rho$ with $\rho \in S_1$ and take $\Pi = \sum_\rho w_\rho \delta_\rho \in L_2$. The equivalence with $\varphi(x) = \infty$ follows from (i). □

Example 10.6 (Each level can fail first)**.** Let $B = \{0, 1\}$, $S_1 = \{\delta_0,\ \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1\}$ and the committed regime $S_2 = \{\delta_{\delta_0}, \delta_{(1/2,1/2)}\}$. Then $L_1 = \{(1-t, t) : 0 \leq t \leq \frac{1}{2}\}$.

(a) *Level one.* $x_1 = \delta_1 \notin L_1$.

(b) *Level two.* $x_2 = \delta_{(3/4,1/4)}$. Its barycenter $(\frac{3}{4}, \frac{1}{4}) = \frac{1}{2}\delta_0 + \frac{1}{2}(\frac{1}{2}, \frac{1}{2})$ lies in L_1, but $(\frac{3}{4}, \frac{1}{4}) \notin S_1$, so $x_2 \notin L_2$. The admissible $\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{(1/2,1/2)}$ has the same barycenter.

(c) *Level three.* $x_3 = \delta_{\Pi'}$ with $\Pi' = \frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{(1/2,1/2)}$. Then $\mu_{\mathsf{D}B}x_3 = \Pi' \in L_2$, but $\Pi' \notin S_2$, so $x_3 \notin L_3$. The admissible $\frac{1}{2}\delta_{\delta_{\delta_0}} + \frac{1}{2}\delta_{\delta_{(1/2,1/2)}}$ has the same barycenter.

In each case the failure is the same move one level up: a mixture of admissible states is asserted as a single state. At level two this is exactly the identity $\frac{1}{2}\delta_{\rho_1} + \frac{1}{2}\delta_{\rho_2} \neq \delta_{(\rho_1+\rho_2)/2}$ behind the failure of the shift law (theorem 8.3).

Proposition 10.7 (Level-three obstructions need third-order data)**.** *In the inherited regime $S_2 = \mathsf{D}(S_1)$ one has $L_3 = \mu_{\mathsf{D}B}^{-1}(L_2)$, so no claim fails first at level three.*

Proof. $\Omega \in \mathsf{D}(\mathsf{D}(S_1))$ iff every $\Pi' \in \operatorname{supp}\Omega$ is supported in S_1, iff

$$\operatorname{supp}\mu_{\mathsf{D}B}(\Omega) = \bigcup_{\Pi' \in \operatorname{supp}\Omega} \operatorname{supp}\Pi' \subseteq S_1. \qquad \square$$

Thus hypothesis support alone produces obstructions only at levels one and two; a genuinely third-order obstruction requires a restriction on which second-order states are admissible.

10.3 Obstruction certificates

We measure obstructions by ℓ^1 distance, $d_k(x_k) = \min_{y \in L_k} \|x_k - y\|_1$, where for $k = 2, 3$ the norm is that of finitely supported measures on $\mathsf{D}^{k-1}B$.

Theorem 10.8 (Level-one certificate)**.** *$x_1 \notin L_1$ if and only if there is $f\colon B \to \mathbb{R}$ with*

$$\textstyle\sum_b f(b)x_1(b) > \max_{\rho \in S_1} \sum_b f(b)\rho(b).$$

Quantitatively,

$$d_1(x_1) = \max_{\|f\|_\infty \le 1} \Big(\langle f, x_1 \rangle - \max_{\rho \in S_1} \langle f, \rho \rangle \Big),$$

and any maximizing f is an optimal certificate.

Proof. L_1 is a polytope in $\mathbb{R}^B$, so the first statement is the separating hyperplane theorem (Farkas' lemma) [39]. For the identity, $d_1(x_1) = \min_{w \in \Delta_{S_1}} \max_{\|f\|_\infty \le 1} \langle f, x_1 - \sum_\rho w_\rho \rho \rangle$, since $\max_{\|f\|_\infty \le 1} \langle f, v \rangle = \|v\|_1$. The function is bilinear and both feasible sets are polytopes, so by the minimax theorem (equivalently LP duality [39]) the min and max may be exchanged, and $\min_w \langle f, \sum_\rho w_\rho \rho \rangle = \max_\rho \langle f, \rho \rangle$ is attained at a vertex. $\square$

Proposition 10.9 (Level-two and level-three certificates)**.** *$x_2 \notin L_2$ iff some $\rho^* \in \operatorname{supp} x_2$ lies outside S_1, and $d_2(x_2) = 2\, x_2(\mathsf{D}B \setminus S_1)$. Likewise $x_3 \notin L_3$ iff some $\Pi^* \in \operatorname{supp} x_3$ lies outside S_2, and $d_3(x_3) = 2\, x_3(\mathsf{D}^2 B \setminus S_2)$. The witnesses ρ^* and Π^* are certificates.*

Proof. For $y \in \mathsf{D}(S)$ and outside mass $m = x(S^c)$, $\|x - y\|_1 \ge \sum_{S^c} x + \big|\sum_S (x - y)\big| = m + |(1 - m) - 1| = 2m$. Equality holds for y equal to x on S with the mass m moved to one point of S. $\square$

Proposition 10.10 (Pooling and obstructions)**.** *For claims $x^{(i)}$ at the same level and weights a_i:*

(i) $d_1(\sum_i a_i x_1^{(i)}) \le \sum_i a_i d_1(x_1^{(i)})$, and the inequality can be strict with every $x_1^{(i)} \notin L_1$ and the pooled claim in L_1;

(ii) $d_k(\sum_i a_i x_k^{(i)}) = \sum_i a_i d_k(x_k^{(i)})$ for $k = 2, 3$.

So pooling can hide level-one obstructions but never level-two or level-three ones.

Proof. (i) d_1 is the distance to a convex set, hence convex. For strictness take $S_1 = \{(\frac{1}{2}, \frac{1}{2})\}$: $d_1(\delta_0) = d_1(\delta_1) = 1$ while $\frac{1}{2}\delta_0 + \frac{1}{2}\delta_1 \in L_1$. (ii) Outside mass is linear in the measure; apply proposition 10.9. In the same example, the level-two claim $\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$ has $d_2 = 2$. $\square$

10.4 Effective admissibility and secondary obstructions

In definition 10.1 every admissible posterior, held with certainty, is an admissible second-order state. This makes each projection $L_{k+1} \to L_k$ surjective (lemma 10.2), so a claim that passes the test at its own level always extends, and the tests of theorem 10.5 are independent membership tests. A mechanism class need not have this property. A class that predicts a fair coin only by mixing two decisive hypotheses realizes the second-order state $\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$ but not certainty in the uniform posterior, $\delta_{(1/2,1/2)}$. We now drop the requirement.

Definition 10.11 (General admissibility data and effective sets)**.** *General admissibility data* are finite sets $S_1 \subseteq \mathsf{D}B$ and $S_2 \subseteq \mathsf{D}(S_1)$, with no further condition. They are *saturated* if $\delta_\rho \in S_2$ for every $\rho \in S_1$, as in definition 10.1. The *local* sets L_1, L_2, L_3 are defined as in definition 10.1, and the *effective* sets are

$$E_3 = L_3, \qquad E_2 = \mu_{\mathsf{D}B}(L_3) = \operatorname{conv}(S_2), \qquad E_1 = \mu_B(E_2) = \operatorname{conv}\big(\mu_B(S_2)\big).$$

Claims and coherent lifts are defined as before, with lifts in $L_1 \times L_2 \times L_3$.

Lemma 10.12. *Each E_k is convex and $E_k \subseteq L_k$. The projections satisfy $\mu_B(L_2) = L_1$, $\mu_{\mathsf{D}B}(L_3) = E_2 \subseteq L_2$ and $\mu_B(E_2) = E_1$. If the data are saturated, then $E_k = L_k$ for all k.*

Proof. $S_2 \subseteq \mathsf{D}(S_1)$ and $\mathsf{D}(S_1)$ is convex, so $E_2 \subseteq L_2$, and $E_1 = \mu_B(E_2) \subseteq \mu_B(L_2) = L_1$. The identities $\mu_B(\mathsf{D}(S_1)) = \operatorname{conv}(S_1)$ and $\mu_{\mathsf{D}B}(\mathsf{D}(S_2)) = \operatorname{conv}(S_2)$ are proved as in lemma 10.2, and $\mu_B(\operatorname{conv} S_2) = \operatorname{conv} \mu_B(S_2)$ because μ_B is affine. If the data are saturated, $\operatorname{conv}(S_2) \supseteq \operatorname{conv}\{\delta_\rho : \rho \in S_1\} = \mathsf{D}(S_1)$. □

Theorem 10.13 (Effective lifting criterion)**.** *Let x be a claim whose highest specified level is m.*

(i) x has a coherent lift if and only if $x_m \in E_m$.

(ii) If $x_m \in L_m \setminus E_m$, then every specified x_k lies in L_k, so x passes every local test of theorem 10.5, yet x has no coherent lift.

(iii) (Certificates) For $x_1 \in L_1$: $x_1 \notin E_1$ if and only if there is $f\colon B \to \mathbb{R}$ with $\langle f, x_1\rangle > \max_{\Pi \in S_2} \langle f, \mu_B \Pi\rangle$, and the distance $d(x_1, E_1)$ satisfies the minimax identity of theorem 10.8 with S_1 replaced by $\mu_B(S_2)$. For $x_2 \in L_2$: $x_2 \notin E_2$ if and only if there is $g\colon S_1 \to \mathbb{R}$ with

$$\textstyle\sum_\rho x_2(\rho) g(\rho) > \max_{\Pi \in S_2} \sum_\rho \Pi(\rho) g(\rho).$$

Proof. (i) If (ρ, Π, Ω) is a lift, then x_m is an iterated barycenter of $\Omega \in L_3$, so $x_m \in E_m$. Conversely, if $m = 3$ take $\Omega = x_3$. If $m \le 2$, write $x_2 = \sum_j w_j \Pi_j$ (respectively $x_1 = \sum_j w_j \mu_B \Pi_j$) with $\Pi_j \in S_2$, and take $\Omega = \sum_j w_j \delta_{\Pi_j} \in L_3$, $\Pi = \mu_{\mathsf{D}B}\Omega = \sum_j w_j \Pi_j \in L_2$ and $\rho = \mu_B \Pi \in L_1$. These agree with the specified levels of x by consistency. (ii) The lower levels of x are iterated barycenters of x_m, and $\mu(L_j) \subseteq L_{j-1}$ by lemma 10.12. Combine with (i). (iii) E_1 and E_2 are convex hulls of finite sets in $\mathbb{R}^B$ and $\mathbb{R}^{S_1}$, so this is the separating hyperplane theorem; the minimax identity is proved as in theorem 10.8. □

The level-two certificate g is a test function on posteriors rather than on outcomes. It detects a claim whose first-order content may be entirely correct but whose second-order structure, meaning which posteriors it asserts and with what confidence, no admissible mechanism produces.

Definition 10.14 (Secondary obstruction)**.** For $x_1 \in L_1$, the *fibre* of level-two lifts is $\mathcal{F}(x_1) = \{\Pi \in L_2 : \mu_B\Pi = x_1\}$, a nonempty compact convex polytope. For $\Pi \in L_2$, the obstruction to extending Π to level three is

$$o_3(\Pi) = \min_{\Pi' \in E_2} \|\Pi - \Pi'\|_1.$$

The *secondary obstruction* of x_1 is the set $I(x_1) = o_3(\mathcal{F}(x_1)) \subseteq [0, \infty)$.

Proposition 10.15 (Indeterminacy)**.** *For $x_1 \in L_1$:*

(i) $I(x_1)$ is a closed interval $[o_3^-(x_1), o_3^+(x_1)]$;

(ii) x_1 has a coherent lift if and only if $0 \in I(x_1)$, if and only if $o_3^-(x_1) = 0$;

(iii) $d(x_1, E_1) \le o_3^-(x_1)$, where d is the ℓ^1 distance;

(iv) o_3^- is convex on L_1;

(v) if the data are saturated, then $I(x_1) = \{0\}$.

Proof. (i) $\mathcal{F}(x_1)$ is compact, convex and nonempty because $\mu_B(L_2) = L_1$, and o_3 is the distance to a convex set, hence continuous and convex. The image of a compact connected set under a continuous function is a compact interval. (ii) E_2 is closed, so $0 \in I(x_1)$ iff some $\Pi \in \mathcal{F}(x_1)$ lies in E_2, iff (by the proof of theorem 10.13) x_1 has a coherent lift. (iii) μ_B is 1-Lipschitz for ℓ^1: $\|\mu_B\Pi - \mu_B\Pi'\|_1 \le \sum_\rho |\Pi(\rho) - \Pi'(\rho)|\,\|\rho\|_1 = \|\Pi - \Pi'\|_1$. Take $\Pi \in \mathcal{F}(x_1)$ attaining $o_3^-(x_1)$ and $\Pi' \in E_2$ nearest to it; then $\mu_B\Pi' \in E_1$. (iv) $(x_1, \Pi) \mapsto o_3(\Pi)$ restricted to the convex set $\{(x_1, \Pi) : \Pi \in L_2,\ \mu_B\Pi = x_1\}$ is jointly convex, and minimizing a jointly convex function over one variable gives a convex function of the other. (v) $E_2 = L_2$ by lemma 10.12, so $o_3 \equiv 0$ on L_2. □

Example 10.16 (Gap, choice dependence and indeterminacy)**.** Let $B = \{0, 1\}$, $u = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and $\Pi_b = \frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$; in all three cases $S_2 = \{\Pi_b\}$, so $E_2 = \{\Pi_b\}$ and $E_1 = \{u\}$.

(a) *Locally admissible, globally non-liftable.* With $S_1 = \{\delta_0, \delta_1\}$, $L_1 = \mathsf{D}B$. The claim $x_1 = (\frac{1}{4}, \frac{3}{4})$ lies in L_1, its only level-two lift is $\frac{1}{4}\delta_{\delta_0} + \frac{3}{4}\delta_{\delta_1}$, and $I(x_1) = \{\frac{1}{2}\} = \{d(x_1, E_1)\}$. The test function $f = (-1, 1)$ certifies $x_1 \notin E_1$ with margin $\frac{1}{2}$.

(b) *Choice dependence.* With $S_1 = \{\delta_0, \delta_1, u\}$, the fibre over $x_1 = u$ is the segment $\Pi_t = \frac{1-t}{2}\delta_{\delta_0} + \frac{1-t}{2}\delta_{\delta_1} + t\,\delta_u$, $t \in [0, 1]$, and $o_3(\Pi_t) = 2t$. So $I(u) = [0, 2]$: the claim u lifts, through $\Pi_0 = \Pi_b$ and $\Omega = \delta_{\Pi_b}$, but the lift $\Pi_1 = \delta_u$ does not extend. As a level-two claim, δ_u passes every local test and is obstructed by the second-order certificate $g = \mathbf{1}_{\{u\}}$: $\langle g, \delta_u\rangle = 1 > 0 = \langle g, \Pi_b\rangle$.

(c) *Nondegenerate indeterminacy without a lift.* With the same data and $x_1 = (\frac{3}{4}, \frac{1}{4})$, the fibre is $\{(\frac{3}{4} - \frac{c}{2})\delta_{\delta_0} + (\frac{1}{4} - \frac{c}{2})\delta_{\delta_1} + c\,\delta_u : 0 \le c \le \frac{1}{2}\}$, $o_3 = \frac{1}{2} + c$ along it, and $I(x_1) = [\frac{1}{2}, 1]$, with $o_3^-(x_1) = \frac{1}{2} = d(x_1, E_1)$.

Case (b) is the phenomenon described above: the first-order claim "fair coin" is correct, while the second-order claim "certainly the uniform posterior" is not, because the class produces fair-coin predictions only through uncertainty between decisive hypotheses. Pooling can hide the secondary obstruction, by proposition 10.15(iv): with $S_1 = \{\delta_0, \delta_1\}$, both δ_0 and δ_1 have $o_3^- = 1$, while their average u has $o_3^- = 0$.

Definition 10.17 (Stratified obstruction)**.** Let x be a claim with highest specified level m. Its *obstruction type* is

(a) *local at level k* if $\varphi(x) = k$, i.e. k is the least level with $x_k \notin L_k$;

(b) *extension-obstructed* if $\varphi(x) = \infty$ but $x_m \notin E_m$; this happens only for $m \in \{1, 2\}$, and for $m = 1$ its secondary obstruction $I(x_1) \subseteq (0, \infty)$ records how far each choice of level-two lift is from extending;

(c) *liftable* otherwise.

For a level-one claim this realizes the scheme of primary and secondary obstructions. Write $o_1(x_1) = d(x_1, L_1)$ for the primary obstruction. Once $o_1 = 0$, a level-two lift always exists, since $\mu_B(L_2) = L_1$, so the level-two obstruction vanishes automatically. The secondary obstruction is the set $I(x_1)$, which is defined only on the fibre of lifts and is said to vanish when $0 \in I(x_1)$. A level-one claim is therefore obstructed at level three precisely when $o_1(x_1) = 0$ and $0 \notin I(x_1)$. For saturated data every claim is local or liftable (proposition 10.15(v)), and theorem 10.5 is recovered.

Remark 10.18 (Relation to classical obstruction theory). The structure parallels obstruction theory for lifting through a tower of fibrations: a primary obstruction, a secondary obstruction defined only once the primary one vanishes, dependence on the choice of intermediate lift, and an indeterminacy whose vanishing (here $0 \in I(x_1)$) is equivalent to the existence of a lift. The differences are substantial. Every level is a convex set, there are no homotopy groups, the obstructions are distances and separating functionals, and the indeterminacy is an interval of values, not a coset of a subgroup. We therefore speak of an *obstruction tower with indeterminacy*, not a Postnikov tower.

10.5 Interpretation and scope

Definition 10.19 (Obstructed claim)**.** A claim is *obstructed* if it is not liftable in the sense of definition 10.17. Its certificate is the test function of theorem 10.8 or the witness state of proposition 10.9 if it is local at level k, and the test function f or g of theorem 10.13(iii) if it is extension-obstructed.

We propose obstructed claims as a precise candidate for what is informally called a *hallucination*: a statement that may look admissible at a lower level but admits no coherent completion through the hierarchy generated by the admissible hypotheses. The definition is exact and computable, since every level reduces to a linear feasibility problem with a dual certificate, and proposition 10.10 suggests testing at levels two and three, where aggregation cannot mask a local failure. The stratification of definition 10.17 distinguishes a claim that is wrong at some level from one that is right at every level separately but cannot be completed coherently, and for level-one claims it records, through $I(x_1)$, how much the outcome depends on which intermediate explanation is chosen. Whether obstruction levels predict errors in trained models is an empirical question that this paper does not address. Section 11 shows the computations on synthetic ensembles, including a valid finite-sample test built from the level-one certificate; it does not test that question. A synthetic level-two experiment and limit theorems are in [9, Sections 51.5 and 52]: a graded transport version of the level-two obstruction ranks post-update errors, and a claim whose atoms lie outside S_1 provably cannot learn [9, Lemma 52.1, Thm. 52.2, Cor. 52.3]. On real data its level-one certificates catch structural violations but not ordinary inaccuracy (same reference, Section 51.5).

Remark 10.20 (Why the obstructions are convex, not homotopical). Each $\mathsf{D}^k B$ is convex and hence contractible, so homotopy-theoretic obstruction classes (such as k-invariants of a Postnikov tower) vanish identically on this tower, and μ has sections, so an unconstrained lift always exists. Obstructions arise only from the admissibility constraints, and they are convex-geometric: separating hyperplanes and support witnesses. The formal shape of primary and secondary obstructions survives nonetheless (remark 10.18).

Remark 10.21 (Relation to contextuality). When B is a product of outcome sets and claims are given as marginals on several contexts, the question whether they come from one admissible joint state is again a linear feasibility problem. This is the setting of sheaf-theoretic contextuality [1], where failure to admit a global section is the obstruction, and of its cohomological witnesses [2], which are sufficient but not necessary for non-extendability. The single-context lifting problem of this section is the base case of that framework. Conversely, a horn of the coupling nerve can be encoded as a level-one claim: the unfillable horns are then exactly the extension-obstructed claims, and the secondary obstruction is twice the contextual fraction when every compatible family is admissible [4, Thm. 11.2]. In that encoding the tower levels do not correspond to horn dimensions [4, Cor. 11.3].

10.6 Sieving as iterated conditioning: a classical illustration

Analytic sieving gives a classical instance of the operations of this paper. We claim no new number theory; standard references are [23, 45]. Let P_0 be uniform on $\{1, \dots, N\}$ and, for a prime p, let $E_p = \{n : p \nmid n\}$.

(a) *Conditioning.* Conditioning on an event E is the information projection [20]

$$P_0(\cdot \mid E) = \operatorname{argmin}\{\mathrm{KL}(P \,\|\, P_0) : P(E) = 1\}.$$

By the chain rule (section 4), conditioning successively on E_p for all primes $p < z$ equals conditioning once on their intersection, the z-rough integers.

(b) *Where the process ends.* If $z^{k+1} > N$, every survivor $n > 1$ has at most k prime factors, and if $z > \sqrt{N}$ the survivors are exactly 1 and the primes in $[z, N]$. Integers with at most k prime factors are therefore intermediate stages, and the primes are the endpoint. For $z = N^{1/u}$ with $u > 1$ fixed, the number of survivors is asymptotic to $u\,\omega(u)N/\log N$, where ω is Buchstab's function [45, III.6]; Dickman's function governs the complementary smooth numbers.

(c) *Parity as observational equivalence.* A linear sieve uses only the data $|\mathcal{A}_d| = \sum_{n \le N,\, d \mid n} w(n)$ for squarefree d up to some level. Selberg observed that the weights $w_\pm(n) = 1 \pm \lambda(n)$, with λ Liouville's function, are supported on integers with an even, respectively odd, number of prime factors, while $|\mathcal{A}_d^\pm| = \lfloor N/d \rfloor \pm \lambda(d) \sum_{m \le N/d} \lambda(m)$, and the last sum is $o(N/d)$. The two weightings are therefore indistinguishable by the sieve data up to errors that sieve methods must tolerate, although they separate parity completely [23, Ch. 16]. This is the situation of theorem 5.2 and proposition 5.7: two different states agree on every statistic the method observes.

(d) *How the barrier is passed.* Chen's theorem that every large even integer is a prime plus a product of at most two primes [19] is consistent with the parity barrier. Passing the barrier, as for primes of the form $x^2 + y^4$ [22], required bilinear ("type II") estimates, that is, statistics beyond the sieve data. This matches proposition 5.7: identifiability is restored by observing more, not by a different geometry on the same observations.

10.7 Support floors, policies and leaf-spring chains

The level-one sets of this section are convex sets of posteriors: L_1, E_1, or any credal set produced by evidence. A decision layer built on them [5, Section 14] adds a *support policy*, a floor $\tau \in \mathbb{R}^B_{\ge 0}$ that a posterior must clear on chosen outcomes, and it reports conclusions as *floors*, the coordinatewise

lower bounds a set certifies. This subsection records how policies and floors interact with the stochastic change of base of corollary 9.15. Write $\mathcal{K}(B)$ for the closed convex subsets of $\mathsf{D}B$, ordered by inclusion; an arrow $L \to L'$ means $L \subseteq L'$, a weakening. For a kernel $k\colon B \to \mathsf{D}(B')$, $k^{\#}$ acts on $\mathcal{K}(B)$ by images and on floors linearly, $(k^{\#}\tau)(b') = \sum_b \tau(b)k(b)(b')$.

Definition 10.22 (Policies and summaries)**.** For a floor τ, the *policy* is $\iota_\tau(L) = L \cap \{\rho \geq \tau\}$. The *floor* of L is $m(L) = \min_{\rho \in L} \rho$, coordinatewise, and its *summary* is the floor box $\sigma(L) = \{\rho \in \mathsf{D}B : \rho \geq m(L)\}$. A set is a *box* if $\sigma(L) = L$.

Proposition 10.23 (Interior and closure)**.** *ι_τ is an interior operator and σ a closure operator on $\mathcal{K}(B)$: both are monotone and idempotent, with $\iota_\tau L \subseteq L \subseteq \sigma L$. Hence, in the poset $\mathcal{K}(B)$ viewed as a category, ι_τ is an idempotent comonad and σ an idempotent monad, and their laws hold trivially.*

Proof. Intersection with a fixed set is deflationary, idempotent and monotone. For σ: $L \subseteq \sigma L$; the floor of the box $\{\rho \geq m\}$ is m (each coordinate attains m_b at the vertex $m + (1 - \sum m)\delta_{b'}$ with $b' \neq b$, or equals m_b when $|B| = 1$), so $\sigma\sigma = \sigma$; and $L \subseteq L'$ gives $m(L) \geq m(L')$, so $\sigma L \subseteq \sigma L'$. In a poset a counit or unit is an inequality and every diagram commutes. □

Proposition 10.24 (Transport of policies and floors)**.** *Let $k\colon B \to \mathsf{D}(B')$ and $h\colon B' \to \mathsf{D}(B'')$ be kernels and $L \in \mathcal{K}(B)$.*

(i) Images compose strictly: $(h \circ k)^{\#}L = h^{\#}k^{\#}L$, so $B \mapsto \mathcal{K}(B)$, $k \mapsto k^{\#}$ is a functor on the Kleisli category, and its category of elements is an opfibration with posetal fibres that contains the discrete opfibration of [4, Prop. 3.3] at level one as the singletons.

(ii) Floors push forward soundly: $\rho \geq \tau$ implies $k^{\#}\rho \geq k^{\#}\tau$; and imposing a policy before transport is stronger than pushing its floor forward: $k^{\#}\iota_\tau L \subseteq \iota_{k^{\#}\tau}(k^{\#}L)$.

(iii) On boxes, $F(k) = \sigma \circ k^{\#}$ is a normal oplax functor into posets: $F(\eta_B) = \mathrm{id}$ and $F(h \circ k)L \subseteq F(h)F(k)L$. The inclusion can be strict.

Proof. (i) $(h \circ k)^{\#} = h^{\#}k^{\#}$ (proposition 9.14(iii)); a category of elements of a functor into posets is an opfibration whose fibres are those posets, and singletons $\{\rho\}$ with $\rho \in A_B^{\mathrm{coh}}$ at level one embed the discrete one. (ii) $k^{\#}$ is linear with non-negative coefficients; if $\rho \in L$ and $\rho \geq \tau$ then $k^{\#}\rho \in k^{\#}L$ and $k^{\#}\rho \geq k^{\#}\tau$. (iii) $\eta_B^{\#} = \mathrm{id}$ and boxes are fixed by σ. Since $k^{\#}L \subseteq \sigma k^{\#}L$ and $h^{\#}$ and σ are monotone, $\sigma h^{\#}k^{\#}L \subseteq \sigma h^{\#}\sigma k^{\#}L$. A strict case is in the example below. □

Definition 10.25 (Leaf-spring chain)**.** A *leaf-spring chain* from (B_0, L_0) is a finite word of steps, each either a *transport* $k_i\colon B_{i-1} \to \mathsf{D}(B_i)$, acting by $k_i^{\#}$, or a *vertical* step at the current base: a policy ι_τ, a summary σ, or a weakening $L \mapsto \mathrm{conv}(L \cup W)$. Its end set is the result of applying the steps in order. A conclusion certified by the chain is a lower bound $\min_{\rho \in L_{\mathrm{end}}} \langle f, \rho \rangle$ for a function f on the final base.

The name records the picture of the proposal behind [5, Section 14]: each segment carries part of the load to the next while flexing locally. Transports are the cocartesian arrows of the opfibration of proposition 10.24(i), vertical steps are arrows (or, for a policy, a coreflection) inside one fibre.

Theorem 10.26 (Leaf-spring chains)**.**

(i) Path independence. *A chain without vertical steps has end set $(k_n \circ \cdots \circ k_1)^{\#}L_0$: it depends only on the composite kernel. Two chains with the same composite and no vertical steps certify exactly the same conclusions.*

(ii) Monotonicity. *Inserting a summary or a weakening anywhere enlarges the end set, so every certified lower bound can only fall; inserting a policy shrinks it, so certified lower bounds can only rise. Certification is lost only at summaries and weakenings, never at a transport.*

(iii) Ordering rule. *Among chains with the same transports, one policy on B_0 and summaries, the summarised end set is smallest when the policy is imposed first and the only summary is taken last. Imposing first is also contained in imposing the pushed floor at the end:* $(k_n \cdots k_1)^{\#}\iota_\tau L_0 \subseteq \iota_{(k_n \cdots k_1)^{\#}\tau}\big((k_n \cdots k_1)^{\#} L_0\big)$.

(iv) Attribution of lost certification. *For a chain with summaries at positions $j_1 < \cdots < j_r$ and a function f, let c_s be the bound certified when the first s summaries are removed. Then $c_0 \le c_1 \le \cdots \le c_r$, and the loss $c_r - c_0 = \sum_s (c_s - c_{s-1})$ is a sum of non-negative contributions, one per summary.*

Proof. (i) is proposition 10.24(i). (ii) All steps are monotone maps of posets, summaries and weakenings are inflationary and policies deflationary, so the end set is monotone in each inserted step. (iii) Moving a policy earlier shrinks the end set by proposition 10.24(ii) applied step by step; removing an intermediate summary shrinks it by (ii), and a final summary is common to all summarised end sets. The displayed inclusion is proposition 10.24(ii) for the composite kernel. (iv) Removing a summary shrinks the end set by (ii), so each $c_s - c_{s-1} \ge 0$; the sum telescopes. □

So the weakest link of a chain is never a transport: it is a vertical step, and the telescoping of (iv) assigns the loss to the summaries. The decomposition depends on the order in which summaries are removed, as any sequential attribution does; a symmetric (Shapley) attribution would average over orders. Path dependence, two routes with the same composite that certify differently, can likewise arise only from the placement of vertical steps. This parallels corollary 9.15: transport never manufactures an obstruction, and here it never loses a certificate; what is lost is lost by summarising.

Example 10.27 (One chain, exactly)**.** Take the kernels $k\colon \{0,1\} \to \mathsf{D}\{u,v,w\}$ and $h\colon \{u,v,w\} \to \mathsf{D}\{P,Q\}$ of [6, Section 11], $k(0) = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, $k(1) = (\frac{1}{5}, \frac{1}{5}, \frac{3}{5})$, $h(u) = \delta_P$, $h(v) = (\frac{1}{2}, \frac{1}{2})$, $h(w) = \delta_Q$, and the credal segment L between $\eta = (\frac{1}{2}, \frac{1}{2})$ and $x_1 = (\frac{7}{10}, \frac{3}{10})$.

- $k^{\#}L$ is the segment from $(\frac{7}{20}, \frac{9}{40}, \frac{17}{40})$ to $(\frac{41}{100}, \frac{47}{200}, \frac{71}{200})$, with floor $(\frac{7}{20}, \frac{9}{40}, \frac{71}{200})$.
- Carried as a set to $\{P,Q\}$, the chain certifies $\rho(Q) \ge \frac{189}{400}$; summarised after k, it certifies only $\frac{187}{400}$. The summary costs $\frac{1}{200}$, and $F(h \circ k) \subsetneq F(h)F(k)$.
- With the policy $\rho(0) \ge \frac{3}{5}$ imposed on $\{0,1\}$ before k, the floor on $\{u,v,w\}$ is $(\frac{19}{50}, \frac{23}{100}, \frac{71}{200})$. Pushing the floor forward instead, $k^{\#}\tau = (\frac{3}{10}, \frac{3}{20}, \frac{3}{20})$, certifies only $(\frac{7}{20}, \frac{9}{40}, \frac{71}{200})$: imposing first gains $\frac{3}{100}$ on u and $\frac{1}{200}$ on v.

11 A synthetic illustration: ensembles, hallucination tests and label noise

This section runs the constructions of section 10 on synthetic data and reads them as statistics and machine learning. It tests nothing about trained models (section 10, Interpretation and scope). Everything is produced by one seeded script, `synthetic_examples.py`, which writes the data as CSV files (section 15). The same script produces the examples of the companion papers [4, 6].

Setting. The base is $B = \{0, 1, 2\}$, three classes. A deep ensemble has $K = 5$ members, each a predictive distribution on B drawn from a Dirichlet law. The members are the admissible posteriors S_1. The admissible second-order states are two ensemble posteriors: uniform weights, and weights $(0.40, 0.30, 0.15, 0.10, 0.05)$ updated on validation data. These form S_2, so the data are not saturated. A *claim* is the predictive distribution that a model reports on an input. Level two is epistemic uncertainty, a law over members; level one is the averaged prediction.

	class 0	class 1	class 2
members $\rho_1, \dots, \rho_5$	$(0.357, 0.086, 0.558)$, $(0.354, 0.314, 0.333)$, $(0.284, 0.478, 0.238)$, $(0.543, 0.080, 0.378)$, $(0.588, 0.100, 0.313)$		
E_1 (uniform, updated)	$(0.425, 0.211, 0.364)$,		$(0.375, 0.213, 0.412)$

Claims. Four claims, with their type (definition 10.17), the ℓ^1 distances to L_1 and E_1, the unmodeled weight $\mu_{\min}$ of theorem 5.10 with the members as sources, and the secondary obstruction $I(x_1)$ of definition 10.14:

claim	x_1	type	$d(x_1, L_1)$	$d(x_1, E_1)$	$\mu_{\min}$	$I(x_1)$
ensemble mean	$(0.425, 0.211, 0.364)$	liftable	0	0	0	$[0, 1.075]$
member mixture	$(0.461, 0.151, 0.387)$	ext.-obstructed	0	0.120	0	$[0.301, 0.944]$
mildly overconfident	$(0.633, 0.091, 0.276)$	local, level one	0.089	0.415	0.114	—
overconfident	$(0.96, 0.03, 0.01)$	local, level one	0.744	1.070	0.958	—

- The *overconfident* claim fails the local test. Its optimal certificate is $f = (1, -1, -1)$, a bet on class 0. Its margin 0.744 equals $d(x_1, L_1)$, as the minimax identity of theorem 10.8 requires. Its unmodeled weight is 0.958: at most 4.2% of the claim can be explained by any sub-mixture of members. This is the contamination reading of theorem 5.10.

- The *member mixture* passes the local test. It is reachable by mixing members with weights $(0.35, 0, 0.15, 0, 0.5)$, but by no admissible ensemble posterior. The model's stated uncertainty cannot come from the ensemble's epistemic state. Its secondary obstruction is the interval $[0.301, 0.944]$: the size of the failure depends on which mixture of members is taken to explain it, and $d(x_1, E_1) = 0.120 \le o_3^-$, as in proposition 10.15(iii).

- The *ensemble mean* lifts, yet $I(x_1) = [0, 1.075]$. Some decompositions of the same prediction are far from admissible, which is the choice dependence of example 10.16(b).

Proposition 11.1 (A finite-sample test from a certificate)**.** *Let $f\colon B \to [-1, 1]$ be fixed independently of an i.i.d. sample $X_1, \dots, X_n \sim x_1$ with empirical law $\hat{x}$. Reject the hypothesis $x_1 \in L_1$ when*

$$\langle f, \hat{x} \rangle - \max_{\rho \in S_1} \langle f, \rho \rangle > \sqrt{2 \log(1/\eta)/n}.$$

If $x_1 \in L_1$, the probability of rejection is at most η. The same holds with S_1 replaced by $\mu_B(S_2)$ and L_1 by E_1.

Proof. If $x_1 \in L_1 = \operatorname{conv} S_1$, then $\langle f, x_1 \rangle \le \max_{\rho \in S_1} \langle f, \rho \rangle$, so rejection requires $\langle f, \hat{x} \rangle - \langle f, x_1 \rangle > t$. Now $\langle f, \hat{x} \rangle$ is a mean of n independent variables with values in $[-1, 1]$ and mean $\langle f, x_1 \rangle$, so by Hoeffding's inequality this has probability at most $\exp(-nt^2/2) = \eta$. □

In practice f is learned on half of the sample by the linear program of theorem 10.8, and the test is applied to the other half. Over 1000 replications at level $\eta = 0.05$:

n (total)	100	200	400	800	1600	3200
null on the boundary of L_1	0.001	0.002	0.002	0.004	0.002	0.003
mildly overconfident ($d = 0.089$)	0.013	0.036	0.075	0.198	0.494	0.875
overconfident ($d = 0.744$)	1.000	1.000	1.000	1.000	1.000	1.000

The test is valid and conservative: its size is far below 0.05, because Hoeffding's bound and the split are loose. Detecting a claim at distance d needs n of order $\log(1/\eta)/d^2$.

Proposition 11.2 (Label noise contracts obstructions)**.** *Let $k\colon B \to \mathsf{D}(B')$ be a kernel with Dobrushin coefficient $\eta(k) = \max_{b,b'} \mathrm{TV}(k(b), k(b'))$. Then*

$$d\big(k^{\#}x_1, \operatorname{conv} k^{\#}S_1\big) \le \eta(k)\, d\big(x_1, \operatorname{conv} S_1\big),$$

and likewise for E_1. For symmetric noise of rate ε on three classes the inequality is an equality, with $\eta = 1 - \frac{3}{2}\varepsilon$.

Proof. For a signed vector v of total mass zero, $\|k^{\#}v\|_1 \le \eta(k)\|v\|_1$ (Dobrushin). Apply this to $v = x_1 - y$ with y a nearest point of $\operatorname{conv} S_1$, and note that $k^{\#}y \in \operatorname{conv} k^{\#}S_1$. For symmetric noise, $k^{\#} = \eta\,\mathrm{id} + (1-\eta)\,u$, where u sends every law to the uniform law. So $k^{\#}$ is a homothety centred at the uniform law, and it scales all distances between laws by η. □

This sharpens corollary 9.15(ii) and (iv), which say only that obstructions cannot appear under transport. For the overconfident claim ($d = 0.744$):

noise	0	0.1	0.2	0.4	0.6	$\frac{2}{3}$	asym. 0.2	asym. 0.5
$\eta(k)$	1	0.85	0.70	0.40	0.10	0	1	1
d after noise	0.744	0.633	0.521	0.298	0.074	0	0.603	0.529

The asymmetric channel flips class 0 to class 1 only; there the bound is not tight. When a model's outputs are scored through noisy labels, the obstruction shrinks with the noise and vanishes at rate $\frac{2}{3}$. So label noise hides hallucinations; it never creates them.

Where the other examples are. The coupling papers apply the same script to a classifier's confusion table. There the Pearson χ^2 statistic is the Fisher form at the product coupling and its degrees of freedom are the dimension of the coupling polytope. Two models scored on one label set form a 2-horn, whose filler polytope bounds joint error metrics, and separately collected tables give horn data whose sampling noise and contextuality separate into primary and secondary obstructions [4, Sections 11–12], [6, Section 15].

12 Nondegeneracy and the likelihood-factorization criterion

Axiom 12.1 (Nondegeneracy)**.** (a) $L(e \mid h_{\mathrm{id}}) = 1$ for all e; (b) $L(e \mid h) > 0$ for all h, e.

Remark 12.2 (Composite identifiability is too strong). The earlier version assumed that $\otimes$ is injective on pairs: $h_1 \otimes h_2 = h_1' \otimes h_2' \Rightarrow (h_1, h_2) = (h_1', h_2')$. Together with (A2) this forces $\mathrm{Hyp}(X) = \{h_{\mathrm{id}}\}$: for any h, $h \otimes h_{\mathrm{id}} = h = h_{\mathrm{id}} \otimes h$, so $(h, h_{\mathrm{id}}) = (h_{\mathrm{id}}, h)$ and $h = h_{\mathrm{id}}$. Restricting to supports does not help when the conclusion is quantified over all posteriors.

Theorem 12.3 (The "iff")**.** *Assume axiom 12.1. Then* $\mathrm{Cond}_e(P_1 \otimes P_2) = \mathrm{Cond}_e(P_1) \otimes \mathrm{Cond}_e(P_2)$ *for all P_1, P_2 (whenever either side is defined) if and only if $L(e \mid h_1 \otimes h_2) = L(e \mid h_1)L(e \mid h_2)$ for all compatible pairs $h_1 \perp h_2$.*

Proof. ($\Leftarrow$) is proposition 4.12. ($\Rightarrow$) Let $h_1 \perp h_2$. If $h_1 = h_2 = h_{\mathrm{id}}$ there is nothing to prove, since $L(e \mid h_{\mathrm{id}}) = 1$. Otherwise fix $x \in X$ and $a, b \in (0,1)$ and set $P_1 = a\delta_{(x,h_1)} + (1-a)\delta_{(x,h_{\mathrm{id}})}$, $P_2 = b\delta_{(x,h_2)} + (1-b)\delta_{(x,h_{\mathrm{id}})}$. Write $\alpha(h_1) = a$, $\alpha(h_{\mathrm{id}}) = 1-a$, $\beta(h_2) = b$, $\beta(h_{\mathrm{id}}) = 1-b$. By (A2) every pair from $\{h_1, h_{\mathrm{id}}\} \times \{h_2, h_{\mathrm{id}}\}$ is compatible, and since all mass sits on one outcome, $P_1 \circledast P_2$ has mass 1. For each $k \in \mathrm{Hyp}(X)$ put

$$w(k) = \sum_{u \otimes v = k} \alpha(u)\beta(v)\, L(e \mid k), \qquad w'(k) = \sum_{u \otimes v = k} \alpha(u)\beta(v)\, L(e \mid u) L(e \mid v),$$

summing over $(u, v) \in \{h_1, h_{\mathrm{id}}\} \times \{h_2, h_{\mathrm{id}}\}$. Then $\mathrm{Cond}_e(P_1 \otimes P_2) = \mathcal{N}(w)$ and, using that all mass is on one outcome, $\mathrm{Cond}_e P_1 \otimes \mathrm{Cond}_e P_2 = \mathcal{N}(w')$. For pairs involving h_{id}, $L(e \mid u \otimes h_{\mathrm{id}}) = L(e \mid u) = L(e \mid u) L(e \mid h_{\mathrm{id}})$ by (A2) and axiom 12.1(a), so

$$w - w' = ab\,\big(L(e \mid h_1 \otimes h_2) - L(e \mid h_1) L(e \mid h_2)\big)\,\delta_{h_1 \otimes h_2}.$$

The composites $h_1 \otimes h_2,\ h_1,\ h_2,\ h_{\mathrm{id}}$ are not all equal (else $h_1 = h_2 = h_{\mathrm{id}}$), so there is $k_0 \neq h_1 \otimes h_2$ among them, and $w(k_0) = w'(k_0) > 0$ by axiom 12.1(b). Equality $\mathcal{N}(w) = \mathcal{N}(w')$ means $w = cw'$ for some $c > 0$; evaluating at k_0 gives $c = 1$, hence $w = w'$ and $L(e \mid h_1 \otimes h_2) = L(e \mid h_1) L(e \mid h_2)$. □

Remark 12.4. No identifiability or separation assumption is needed: the hypothesis h_{id} already supplies a reference point that fixes the normalizing constant. The earlier proof's constant $c = Z/(Z_1 Z_2)$ depended on the chosen posteriors and could not be set to 1 by a separate choice.

13 Coalgebras, bialgebras and the 2-categorical setting

This section describes the coalgebras, the lifted monad and the bialgebras of the base-relative law λ^B of theorem 9.8, and makes precise which 2-category is involved.

13.1 The ambient 2-category

The monad D and the comonad C^B are a monad and a comonad on the object **Set** of the 2-category **Cat** of categories, functors and natural transformations, in the sense of the formal theory of monads [43]. Their structure maps $\eta, \mu, \varepsilon, \delta$ and the law λ^B are 2-cells of **Cat**, namely natural transformations between endofunctors of **Set**, and these 2-cells are not identities. Theorem 9.8 is therefore a statement in **Cat**, and by the correspondence of [38] it is equivalent to a lifting of D to the category of C^B-coalgebras.

The categories of coalgebras, algebras and bialgebras obtained in this way are ordinary categories. Regarded as 2-categories (for instance by viewing each set as a discrete category), they are *locally discrete*: a natural transformation between functors of discrete categories exists only between equal functors, so the only 2-cells are identities. Section 13.5 draws the consequences.

13.2 Coalgebras of the base-relative comonad

Proposition 13.1. *A map $\gamma\colon Y \to Y \times A_B$ is a C^B-coalgebra structure if and only if $\gamma = \langle \mathrm{id}_Y, g\rangle$ for a (unique) map $g\colon Y \to A_B$, and a map $f\colon Y \to Y'$ is a coalgebra morphism $(Y, \langle \mathrm{id}, g\rangle) \to (Y', \langle \mathrm{id}, g'\rangle)$ if and only if $g' \circ f = g$. Hence $\mathrm{Coalg}(\mathsf{C}^B) \cong \mathbf{Set}/A_B$, and likewise $\mathrm{Coalg}(\mathsf{C}^B_{\mathrm{coh}}) \cong \mathbf{Set}/A_B^{\mathrm{coh}}$.*

Proof. The counit law $\varepsilon_Y \circ \gamma = \mathrm{id}_Y$ forces the first component of γ to be the identity, so $\gamma = \langle \mathrm{id}, g\rangle$. Coassociativity then holds automatically: both $\mathsf{C}^B(\gamma) \circ \gamma$ and $\delta_Y \circ \gamma$ send y to $((y, g(y)), g(y))$. The condition $(f \times \mathrm{id}) \circ \gamma = \gamma' \circ f$ reads $(f(y), g(y)) = (f(y), g'(f(y)))$. □

Proposition 13.2 (Cofree coalgebras)**.** *The forgetful functor* $U\colon \mathrm{Coalg}(\mathsf{C}^B) \to \mathbf{Set}$ *has a right adjoint* F*, with* $F(X) = (X \times A_B, \delta_X)$*, i.e.* $X \times A_B$ *over* A_B *by* pr_2*. Coalgebra morphisms* $h\colon (Y,\gamma) \to F(X)$ *correspond to maps* $\varphi\colon Y \to X$ *by* $h \mapsto \mathrm{pr}_1 \circ h$ *and* $\varphi \mapsto (\varphi \times \mathrm{id}) \circ \gamma = \langle \varphi, g\rangle$*. The counit of* $U \dashv F$ *at a set* X *is* $\varepsilon_X = \mathrm{pr}_1\colon X \times A_B \to X$*, and the unit at a coalgebra* (Y,γ) *is the structure map* $\gamma = \langle \mathrm{id}_Y, g\rangle\colon (Y,\gamma) \to FU(Y,\gamma)$*.*

Proof. Under proposition 13.1, a morphism $(Y,g) \to (X \times A_B, \mathrm{pr}_2)$ over A_B is a map $h = \langle \varphi, g\rangle$, determined by $\varphi = \mathrm{pr}_1 \circ h$. The unit is the morphism corresponding to id_Y, which is $\langle \mathrm{id}_Y, g\rangle$. □

Remark 13.3. For a bare set X there is no natural map $X \to X \times A_B$: a map $x \mapsto (x, e)$ requires a choice of $e \in A_B$, and it is a coalgebra morphism only out of the constant coalgebra (X, const_e). The unit of the cofree adjunction is the coalgebra structure itself, as in proposition 13.2.

13.3 The lifted monad

Proposition 13.4. *Under the isomorphism of proposition 13.1, the lifting of* D *determined by* λ^B *is*

$$\tilde{\mathsf{D}}(Y,g) = \big(\mathsf{D}Y,\ \beta_B \circ \mathsf{D}(g)\big), \qquad \tilde{\mathsf{D}}(f) = \mathsf{D}(f),$$

with unit and multiplication those of D*. Conversely,* λ^B_X *is the coalgebra morphism* $\tilde{\mathsf{D}}(FX) \to F(\mathsf{D}X)$ *corresponding under proposition 13.2 to* $\mathsf{D}(\varepsilon_X) = \mathsf{D}(\mathrm{pr}_1)$*.*

Proof. The lifting induced by a mixed law sends a coalgebra (Y,γ) to $(\mathsf{D}Y, \lambda^B_Y \circ \mathsf{D}\gamma)$ [38]. For $\gamma = \langle \mathrm{id}, g\rangle$ and $P \in \mathsf{D}Y$, $\lambda^B_Y(\mathsf{D}\gamma(P)) = (P, \beta_B(\mathsf{D}g(P)))$, which is the coalgebra $\langle \mathrm{id}, \beta_B \circ \mathsf{D}g\rangle$. That η_Y and μ_Y are maps over A_B is the content of (U) and (M); directly, $\beta_B(\mathsf{D}g(\delta_y)) = g(y)$ by the unit law of the algebra β_B, and $\beta_B \circ \mathsf{D}g \circ \mu_Y = \beta_B \circ \mu_{A_B} \circ \mathsf{D}^2 g = \beta_B \circ \mathsf{D}(\beta_B \circ \mathsf{D}g)$ by naturality of μ and the associativity law of β_B. For the converse, $\tilde{\mathsf{D}}(FX)$ is $\mathsf{D}(X \times A_B)$ over A_B by $\beta_B \circ \mathsf{D}(\mathrm{pr}_2)$, so by proposition 13.2 the morphism corresponding to $\mathsf{D}(\mathrm{pr}_1)$ is $\langle \mathsf{D}(\mathrm{pr}_1), \beta_B \circ \mathsf{D}(\mathrm{pr}_2)\rangle = \lambda^B_X$. □

13.4 Bialgebras

Recall that a λ-bialgebra is a set Y with a D-algebra structure $a\colon \mathsf{D}Y \to Y$ and a C^B-coalgebra structure γ such that $\gamma \circ a = \mathsf{C}^B(a) \circ \lambda_Y \circ \mathsf{D}(\gamma)$. Algebras of D are convex spaces [24], and their morphisms are affine maps.

Proposition 13.5. *A* λ^B*-bialgebra is exactly a convex space* (Y,a) *together with an affine map* $g\colon (Y,a) \to (A_B, \beta_B)$*. Hence the category of* λ^B*-bialgebras is isomorphic to the slice of the category of convex spaces over* (A_B, β_B)*, and the coherent variant to the slice over* $(A_B^{\mathrm{coh}}, \beta_B)$*.*

Proof. With $\gamma = \langle \mathrm{id}, g\rangle$ and $P \in \mathsf{D}Y$, the right side of the bialgebra condition is

$$\mathsf{C}^B(a)\big(P, \beta_B(\mathsf{D}g(P))\big) = \big(a(P), \beta_B(\mathsf{D}g(P))\big),$$

and the left side is $(a(P), g(a(P)))$. So the condition is $g \circ a = \beta_B \circ \mathsf{D}(g)$, which says that g is a morphism of D-algebras. □

Example 13.6. By proposition 9.5, $(\mathrm{Sh}(B), \mu_{\mathrm{Hyp}B}, \sigma_B)$ is a bialgebra with values in A_B^{coh}: the space of shadows, mixed as distributions, together with the canonical hierarchy it determines. For any coalgebra (Y,g), the free bialgebra is $(\mathsf{D}Y, \mu_Y, \beta_B \circ \mathsf{D}g)$. A map that reads a context off a state non-affinely, for instance from the first hypothesis in the support of a shadow, violates the bialgebra condition.

So a bialgebra is a convex space of states with an affine readout of a three-level context: mixing states mixes their contexts level by level.

13.5 What two-dimensional monad theory does and does not add

Because the categories of propositions 13.1, 13.4 and 13.5 are locally discrete as 2-categories, every 2-cell in them, invertible or not, is an identity. Consequently, for a 2-monad on such a 2-category, pseudo-algebras and pseudo-morphisms are strict ones, lax morphisms are strict ones, and the inclusion of strict algebras into pseudo-algebras is an isomorphism. The coherence and strictification results of Blackwell–Kelly–Power [11, 36], their notion of flexibility, two-dimensional monadicity [12] and doctrinal adjunction [35] therefore reduce to statements about ordinary categories and give no information beyond theorem 9.8 and propositions 13.1, 13.2, 13.4 and 13.5. We make no flexibility, strictification or 2-monadicity claims.

Genuine two-dimensional content would require three things that are not available in this paper:

(i) an ambient 2-category in which contexts have non-identity morphisms, with specified hom-categories;

(ii) a 2-monad extending D to that 2-category, with a specified action on 2-cells, and a comonad lifted along it;

(iii) verification, for that 2-monad, of the codescent and doctrinal-adjunction hypotheses on which the results cited above depend.

13.6 Convex-order Beck comparisons for the absolute tower

Since the absolute tower admits no strict law (theorems 8.3 and 8.7), it is natural to ask whether an order on the relevant hom-sets turns its Beck defects into 2-cells. We test the most natural candidate, the convex order, and show that it fails in two independent ways. Throughout we write *convex-order Beck comparison* for an inequality between the two sides of a Beck diagram; no lax or oplax distributive law is claimed.

Definition 13.7 (Convex order)**.** Let K be a convex subset of a real vector space. For finitely supported $\Pi, \Pi' \in \mathsf{D}(K)$ write $\Pi \leq_{\mathrm{cx}} \Pi'$ if $\sum_k \Pi(k)\varphi(k) \leq \sum_k \Pi'(k)\varphi(k)$ for every convex $\varphi\colon K \to \mathbb{R}$. Then Π and Π' have the same barycenter, and by Strassen's theorem [42] $\Pi \leq_{\mathrm{cx}} \Pi'$ if and only if there is a *martingale coupling*: a coupling π of Π and Π' such that, for each atom k of Π, the conditional distribution $\pi(k,\cdot)/\Pi(k)$ has barycenter k.

We apply this with $K = \mathsf{D}X$ (a simplex in $\mathbb{R}^X$) and $K = \mathsf{C}_3(\mathsf{D}X) = \mathsf{D}X \times \mathsf{E}_3(\mathsf{D}X)$, which is convex as a product of convex sets, each $\mathsf{D}(Z)$ sitting affinely in the vector space of finitely supported functions on Z. Composites of convex functions with affine maps are convex.

Proposition 13.8 (Unit and multiplication comparisons)**.** *For the shift law λ^σ of theorem 8.3, on level-one components:*

(a) (unit) the level one of $\lambda^\sigma \eta_{\mathsf{C}_3}(x,c)$ is δ_ρ, that of $\mathsf{C}_3(\eta_X)(x,c)$ is $\mathsf{D}(\eta_X)\rho$, and $\delta_\rho \leq_{\mathrm{cx}} \mathsf{D}(\eta_X)\rho$;

(b) (multiplication) with M' as in the proof of theorem 8.3, the level one of $\mathsf{C}_3(\mu_X)\lambda^\sigma_{\mathsf{D}X}\mathsf{D}(\lambda^\sigma_X)(M')$ is $\sum_\ell \nu_\ell \delta_{\bar\rho_\ell}$ with $\bar\rho_\ell = \sum_k a_{\ell k}\rho_{\ell k}$, that of $\lambda^\sigma_X \mu_{\mathsf{C}_3 X}(M')$ is $\sum_{\ell,k} \nu_\ell a_{\ell k}\delta_{\rho_{\ell k}}$, and the former is $\leq_{\mathrm{cx}}$ the latter.

Both inequalities are strict (not equalities) as soon as ρ is not a point mass, respectively some group $\{\rho_{\ell k}\}_k$ has two distinct members.

Proof. (a) The coupling $\pi(\rho, \delta_x) = \rho(x)$ is a martingale coupling, since $\sum_x \rho(x)\delta_x = \rho$. (b) The coupling $\pi(\bar{\rho}_\ell, \rho_{\ell k}) \mathrel{+}= \nu_\ell a_{\ell k}$ is a martingale coupling, since the conditional barycenter of $\bar{\rho}_\ell$ is $\sum_k a_{\ell k}\rho_{\ell k}$. Strictness: for the strictly convex $\varphi(v) = \|v\|_2^2$, Jensen's inequality is strict exactly when the averaged points are not all equal. □

Proposition 13.9 (Orientation obstruction)**.** *Write each Beck axiom of definition 2.1 as an equation between its λ-*first side *($\lambda \circ \eta_C$, respectively $\lambda \circ \mu_C$) and its other side ($C\eta$, respectively $C\mu \circ \lambda_T \circ T\lambda$). In proposition 13.8 the λ-first side is* below *in the unit comparison and* above *in the multiplication comparison, and the reverse inequalities fail. Hence no choice of orientation of $\leq_{\mathrm{cx}}$ makes both comparisons 2-cells of the same kind: reading $\leq_{\mathrm{cx}}$ as a 2-cell, the unit comparison has the lax shape and the multiplication comparison the oplax shape, and reversing the order exchanges them.*

Proof. The orientations are those of proposition 13.8; the reverse inequalities fail by its strictness statement. □

The two comparisons are nonetheless consistent with each other: taking $M' = \delta_M$, the unit comparison at $\mathsf{D}X$ and monotonicity of $\mathsf{D}(\mu_X)$ (pushforward along an affine map preserves $\leq_{\mathrm{cx}}$) give $\mathsf{C}_3(\mu_X)\lambda^\sigma_{\mathsf{D}X}(\delta_{\lambda^\sigma M}) \leq_{\mathrm{cx}} \lambda^\sigma(M)$ at level one, in agreement with proposition 13.8(b).

Proposition 13.10 (Comparability obstruction)**.** *For both the shift law λ^σ and the averaging law $\bar{\lambda}$, take $X = \{0, 1\}$ and $M = \frac{1}{2}\delta_{(0,c_0)} + \frac{1}{2}\delta_{(0,c_1)}$ with c_i differing only in $\rho_i = \delta_i$. At level one, the two sides of (Cm) are*

$$L = \tfrac{1}{2}\delta_{(\delta_0,e)} + \tfrac{1}{2}\delta_{(\delta_1,e)}, \qquad R = \tfrac{1}{2}\delta_{(\delta_0,e_0)} + \tfrac{1}{2}\delta_{(\delta_1,e_1)} \qquad \text{in } \mathsf{D}(\mathsf{C}_3\mathsf{D}X),$$

where e_i are the per-sample contexts and $e = \frac{1}{2}e_0 + \frac{1}{2}e_1$ is the pooled context. They have the same barycenter, and they are incomparable: $L \not\leq_{\mathrm{cx}} R$ and $R \not\leq_{\mathrm{cx}} L$.

Proof. The forms of L and R are computed in the proofs of theorems 8.3 and 8.7. Let $s(P) = P(1)$ on $\mathsf{D}X$ and let $\tau(e')$ be the mass that the level-one component of e' assigns to δ_1; both are affine, and $\tau(e_0) = 0$, $\tau(e_1) = 1$, $\tau(e) = \frac{1}{2}$. The functions

$$\varphi_1(P, e') = \big(s(P) - \tau(e')\big)^2, \qquad \varphi_2(P, e') = \big(\tau(e') - \tfrac{1}{2}\big)^2$$

are convex. Then $\sum L\varphi_1 = \frac{1}{4} > 0 = \sum R\varphi_1$, so $L \not\leq_{\mathrm{cx}} R$, and $\sum R\varphi_2 = \frac{1}{4} > 0 = \sum L\varphi_2$, so $R \not\leq_{\mathrm{cx}} L$. Equivalently, in Strassen's terms: a martingale coupling must keep the first coordinate of each atom, which pins (δ_0, e) to (δ_0, e_0) and (δ_0, e_0) to (δ_0, e), contradicting the second coordinate. □

Corollary 13.11. *The naive convex-order relaxation of the absolute tower fails: the convex order orients the unit and multiplication defects of λ^σ oppositely (proposition 13.9) and cannot compare the comultiplication defect of either candidate law (proposition 13.10). In particular, neither λ^σ nor $\bar{\lambda}$ becomes a lax or oplax mixed distributive law by reading $\leq_{\mathrm{cx}}$ as 2-cells.*

The first version of the paper proposed a "barycentric 2-cell" for the multiplication defect. Proposition 13.8(b) is the correct form of that idea, an order relation and not a map in **Set**, and corollary 13.11 shows why it does not extend to a law.

Remark 13.12 (Toward a lifting order). The comparability obstruction has a clear source. The convex order compares only barycentric spreads, whereas the two sides of (Cm) differ in how contexts are attached to samples: one pooled context versus one context per sample, with equal barycenters. An enrichment that can orient this defect must compare the hierarchical coherence data of section 10, namely whether and how a coherent lift through the tower exists, and not only barycenters. One

natural attempt fails and shows what is needed. Call $x \preceq x'$ if every admissibility datum (S_1, S_2) that lifts x' also lifts x. At level one this is equality: if $x \preceq x'$, take $S_1 = \{x'\}$, so $L_1 = \{x'\}$ and $x = x'$. A useful *lifting order* must therefore be relative to a fixed admissibility datum or carry additional structure, such as the certificates of theorem 10.8 and proposition 10.9, the effective sets of definition 10.11, or the secondary obstruction of definition 10.14. Constructing one that orients all three Beck defects of the absolute tower in the same direction is left open.

14 Empirical setup

The empirical setting is the CausalIDView benchmark of Jung, Seo, Byun, Lee and Song [34], which evaluates causal models across observational views; CausalIDView itself is not gradient-free. The method evaluated here is ours: a gradient-free treatment with structured shadows, developed in [9, Section 51]. The outcome space X consists of binary causal identification outcomes; descriptors θ are sparse multi-index hypotheses with spline links, nominally $r = 3$ directions; the posterior is sampled by gradient-free Metropolis. Targets are Manski-type partial-identification bounds [37] and instrumental-variable bounds.

On the 40-world benchmark (appendix A), the posterior under a generic prior with 16 pooled chains attains Manski RMSE 0.0934 and IV RMSE 0.1226, against 0.105 and 0.138 for the previous structured backbone. These numbers are close to the approximate TabPFN reference values [31]; since TabPFN was not rerun under the same protocol and no uncertainty intervals are reported, we do not claim parity. The flattening of the chain-pooling curve is consistent with, but does not by itself test, saturation of the level-two hierarchy. The 90% bootstrap intervals of the structured backbone cover the true bounds only 70–71% of the time, which indicates substantial miscalibration of that backbone. Coverage of the intervals produced by the generic-prior posterior has not been measured; bias-aware intervals remain open.

The categorical results above do not depend on these experiments, and the experiments do not test the categorical results.

15 Code and data availability

The verification scripts and related material are available at the repository below. The scripts are:

- `check_tower.py`: the absolute tower;
- `check_base_relative.py`: the base-relative tower;
- `check_lifting.py`: lifting and obstruction certificates;
- `check_obstruction_tower.py`: effective admissibility and secondary obstructions;
- `check_context_decomposition.py`: identifiability, contamination and sieving;
- `check_coalgebras.py`: coalgebras and bialgebras;
- `check_convex_order.py`: convex-order Beck comparisons;
- `check_stochastic_basechange.py`: stochastic change of base, and the toric geometry of coupling polytopes used in [3, 4];
- `synthetic_examples.py`: the synthetic data of section 11, of [4, Section 12] and of [6, Section 15]. It is seeded, runs in about twenty seconds and checks its own claims (8 checks). The generated files are also archived at https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/blob/main/coupling/synthetic_data.zip. It writes `results.json` and nine CSV files:
  - `triage_samples`, `chamber_identification`, `independence_tests`;
  - `two_models_samples`, `horn_studies`;
  - `ensemble_members`, `ensemble_claims`, `hallucination_test`, `label_noise`.

The scripts for the coupling nerve, the toric geometry and the tower-and-nerve encoding are listed in [4, Section 14].

https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/Heirarchical_posteriors

16 Related work

Three-layer realisation, hierarchy and holonomy. [9, 7, 8].

Categorical probability. The finite distribution monad and Markov categories [18, 26]; a synthetic de Finetti theorem [25].

Distributive laws. Beck's distributive laws [15]; the formal theory of monads [43]; mixed laws between a monad and a comonad and their correspondence with liftings [38]; weak distributive laws [44, 16, 17], with applications to the Vietoris monad [28] and to combining probabilistic and nondeterministic choice [29].

Couplings, the coupling nerve and toric geometry. The 2-cells of the Kleisli category, the coupling nerve, cumulants and the toric geometry of coupling polytopes, and their relation to the obstructions of section 10, are developed in [4], with computations in [6].

Convex duality. Farkas' lemma, LP duality and the minimax identity used for obstruction certificates [39].

Information cohomology and contextuality. Work of Baudot and Bennequin [14], Vigneaux [46], Abramsky–Brandenburger [1] and Abramsky, Mansfield and Barbosa [2].

Coalgebras and two-dimensional monad theory. Convex spaces as algebras of D [24]; two-dimensional monad theory [11, 36], two-dimensional monadicity [12] and doctrinal adjunction [35]; the convex order and Strassen's theorem [42].

Empirical. The CausalIDView benchmark [34]; the gradient-free method applied to it [9, Section 51]; TabPFN [31]; partial identification [37]; spiked random-matrix thresholds [13].

17 Summary of results

1. Base-relative tower: coherent contexts form a sub-algebra; the canonical hierarchy is an affine, natural section; λ^B is a strict mixed distributive law satisfying all four Beck axioms, natural in the base; pooling commutes with hierarchy formation and with conditioning (proved).

2. Coherent lifting: admissibility is convex at each level, failure is monotone, each level can fail first, level-three failures need third-order data, every failure has a certificate, and pooling hides level-one but not level-two or level-three obstructions (proved).

3. Obstruction tower with indeterminacy: for unsaturated data, lifts exist exactly on the effective sets, with first- and second-order certificates; the secondary obstruction on the fibre of level-two lifts depends on the chosen lift, ranges over an interval containing zero exactly when a lift exists, is bounded below by the effective distance, and can be hidden by pooling (proved, with worked examples).

4. Identifiability: source weights are unique for every mixture iff the sources are affinely independent; exclusive regions identify individual weights; supports-only information gives the core of a belief function, and for two sources the sharp interval $[x(R_A), 1 - x(R_B)]$; under unmodeled contamination the data certify a lower bound on the unmodeled weight, the distance test is one-sided, and the number of unmodeled sources is not identifiable (proved).

5. Coalgebras of the base-relative comonad are sets over A_B, the lifted monad is $(Y, g) \mapsto (\mathsf{D}Y, \beta_B \circ \mathsf{D}g)$, and bialgebras are convex spaces with an affine context map (proved). The ambient 2-category is **Cat**; the resulting categories are locally discrete, so two-dimensional monad theory adds nothing here. The naive convex-order relaxation of the absolute tower fails: unit and multiplication comparisons have opposite orientations and the comultiplication defect is incomparable (proved, with exact certificates).

6. The absolute tower grows to $\mathsf{D}^4 X$ under D (remark 6.5).

7. $\mathsf{C}_\mathsf{E}(X) = X \times \mathsf{E}(X)$ is a comonad for every E; C_3 is an instance (proved).

8. (Hyp, q) with q natural; $J = \mathrm{Hyp}(j)$ satisfies CL1–CL3; $K^{[k]}$ is natural and coheres with J via $\mathsf{D}^3 j$ (proved).

9. Normalized partial algebra on $\mathrm{Post}(X)$ with reweighted distributivity and conditioning laws (proved).

10. Affine first isomorphism theorem; congruence for $\otimes$ when $\perp$ is total; congruence for Cond_e iff $L(e \mid \cdot)$ is constant (proved, with counterexamples).

11. Absolute tower: shift law natural and (Co), fails (U), (M), (Cm); averaging law natural, (U), (Co), (M), fails (Cm). The unit axiom excludes point-mass hierarchy at level one (proved).

12. Composite identifiability trivializes Hyp; the "iff" holds under nondegeneracy alone (proved).

13. Support floors: policies are interior and summaries closure operators on level-one credal sets; floors push forward soundly, policies imposed before transport certify more, and floor summaries form a normal oplax functor; along a leaf-spring chain certification is path-independent and is lost only at summaries, with an exact attribution (proved, with an exact example).

18 Open problems

1. For the absolute tower, does any mixed distributive law $\mathsf{DC}_3 \Rightarrow \mathsf{C}_3\mathsf{D}$ satisfy all four axioms of definition 2.1? (For the base-relative tower this is settled by theorem 9.8.)

2. Bayesian updating as a morphism: make Cond_e of proposition 9.12 a natural operation on $\mathsf{DC}^B_{\mathrm{coh}}$ compatible with λ^B, including evidence on outcomes.

3. Is there an enrichment in which all Beck defects of λ^σ or $\bar{\lambda}$ are witnessed by 2-cells of one orientation, giving a genuine lax or oplax law? The convex order does not work (corollary 13.11); remark 13.12 suggests comparing coherent lifts instead, and shows that the naive liftability preorder degenerates.

4. Specify a 2-category with non-identity 2-cells and a 2-monad extending D to it, as in section 13.5, and determine whether flexibility, strictification or 2-monadicity results then say anything about the coherent context algebra.

5. Weaken axiom 12.1 in theorem 12.3, e.g. allow zero likelihoods.

6. Secondary obstructions for higher-level claims and for restricted level-one data, and an efficient characterization of the upper end o_3^+ of the indeterminacy interval, which maximizes a convex function over the fibre.

7. Contamination at levels two and three: extend theorem 5.10 to unmodeled second-order states, and characterize the least number of unmodeled sources when they are restricted to a family not closed under mixing.

8. Multi-context obstruction towers: combine section 10.4 with the sheaf-theoretic setting of contextuality, where cohomological witnesses [2] could supply obstructions of a genuinely cohomological kind.

9. Approximate admissibility: replace exact support conditions at levels two and three by transport distances on $\mathsf{D}B$ and D^2B, whose Kantorovich duals would give graded certificates. The level-two case is done in [9, Lemma 52.1]; level three is open.

10. Empirical test: compute failure levels and certificates for model claims with known errors, and determine whether level-two and level-three obstructions predict them. Partly answered on synthetic data in [9, Sections 51.5 and 52]; a real-data level-two test is open.

11. Extension to non-discrete outcome spaces (Giry monad).

12. Calibrated uncertainty and a test that could falsify the level-two saturation interpretation of the chain-pooling curve.

19 Conclusion

Keeping exactly three posterior levels requires fixing the base on which they live. In the absolute tower the levels are re-indexed by the object, so applying the monad raises the order to D^4X, and neither natural candidate law satisfies all Beck axioms; the unit axiom alone rules out point-mass hierarchy at level one. In the base-relative tower the levels stay over a fixed outcome base B, coherent contexts form a sub-algebra of the monad, the canonical hierarchy ρ_h, δ_{ρ_h}, $\delta_{\delta_{\rho_h}}$ is an affine section, and pooling defines a strict mixed distributive law satisfying all four Beck axioms. Pooling commutes with hierarchy formation and, after evidence reweighting, with conditioning, while distinct posteriors are retained at level two. The same tower supports an exact lifting problem: claims are checked level by level against the states a hypothesis class can produce, every failure comes with a certificate, and obstructions above level one survive aggregation. When the admissible second-order states are restricted, the lifting problem acquires a secondary obstruction that depends on the choice of intermediate lift, and a claim can be correct at every level separately yet admit no coherent completion. Attribution of a mixed context to its sources is exact for affinely independent

sources, sharply bounded in the other regimes, and, under unmodeled contamination, complete only relative to a declared list of sources and a declared budget: the data certify how much is unexplained, but not how many unmodeled mechanisms there are. This makes "hallucination as obstructed lifting" a precise, testable definition. The coalgebras and bialgebras of the base-relative law are concrete, namely sets and convex spaces over the context algebra, and genuine two-dimensional structure is not present in this setting. For the absolute tower, relaxing the Beck equations to the convex order does not help: the unit and multiplication comparisons point in opposite directions and the comultiplication defect cannot be compared. The base-relative theorem therefore remains the control result, and the open question is which enrichment that compares coherent lifts, rather than barycentric spreads, could orient all the defects at once.

A Empirical tables

A.1 The 40-world benchmark

All numbers are on the 40-world set. The structured-backbone row uses 30 bootstrap refits per world; the posterior rows are computed from pooled Metropolis chains (8 or 16 chains), not from bootstrap refits. The protocols for the remaining rows are listed for confirmation in appendix B.

Method	Manski RMSE	IV RMSE
Logistic regression (rebuild)	0.132	0.169
Structured backbone (previous best)	0.105	0.138
Posterior under generic prior, 8 chains	0.0937	0.1230
Posterior under generic prior, 16 chains	0.0934	0.1226
TabPFN, approximate reference[†]	≈ 0.09	≈ 0.12
Constant predictor	0.131	—

[†]Approximate values, not produced under this benchmark's protocol; see appendix B, item 13.

A.2 Chain-pooling curve

Chains	1	2	4	8	16
Manski	0.1005	0.0973	0.0953	0.0937	0.0934
IV	0.1319	0.1277	0.1250	0.1230	0.1226

Differences between 8 and 16 chains (about 3×10^{-4}) are reported without standard errors and should not be over-interpreted.

A.3 Coverage

The 90% bootstrap intervals of the structured backbone cover the true bounds 70–71% of the time (82% before the more regularized backbone). Coverage of the generic-prior posterior's intervals has not been measured.

A.4 Shallow-MLP worlds

On shallow-MLP worlds the generic-prior posterior scores 0.104 against 0.103 for the structured backbone, i.e. no improvement. A misspecified level-two shadow is one possible explanation; it has not been tested.

A.5 Spectral pre-pass

The pre-pass recovers on average 2.5 of 6.6 planted directions and 4.5 of 12 active coordinates per direction. In spiked random-matrix models, the Baik–Ben Arous–Péché transition [13] gives a signal-strength threshold below which the top sample eigenvector carries no information about a planted direction. Whether our spectral pre-pass operates in such a regime has not been established, so we state the low recovery rate without attributing it to that threshold.

B Changes from earlier versions

Addition to version 10 (October 2026). New section 10.7: support policies and floor summaries on level-one credal sets, their transport along kernels, and leaf-spring chains (propositions 10.23 and 10.24 and theorem 10.26, example 10.27); new script `check_support_floors.py`. It is the last subsection of section 10, so no earlier or later numbering changes.

Version 10.

(a) New section 11: a synthetic deep-ensemble illustration of the lifting problem, a finite-sample test built from a certificate (proposition 11.1), and contraction of obstructions under label noise (proposition 11.2); new script `synthetic_examples.py` with its CSV data.

(b) Cross-references to the coupling papers [4, 6]: Kleisli 2-cells and the opfibration (section 9.7), the horn encoding (Relation to contextuality), related work, and code.

(c) Numbering of section 10 and earlier is unchanged; later sections move up by one.

(d) Open problems 9 and 10 now point to their partial answers in [9].

Version 9.

(a) New section 9.7: change of base along Markov kernels (proposition 9.14), transport of admissibility data, lifts, certificates and secondary obstructions (corollary 9.15), and conditioning through a channel (proposition 9.16). This closes a gap noted after version 8: lemma 9.3 and proposition 9.5 were established only for deterministic base change.

(b) A remark relating theorem 5.10 to mixture proportion estimation and to contaminated sampling [10, 40, 30]; new references [3, 33]; new script `check_stochastic_basechange.py`.

(c) The publicly posted earlier version (DOI 10.13140/RG.2.2.29743.50088) carries the title "... and Weak Distributive Laws". The weak-law claim was withdrawn in a later revision (remark 8.9), and the title now reads "Mixed Distributive Laws".

Version 8.

(a) New section 5.2: identifiability of source weights (proposition 5.7), the supports-only interval and belief-function core (proposition 5.8), and a summary of the three regimes.

(b) New section 5.3: the contamination model, the certified lower bound on the unmodeled weight with its dual certificate and budgeted bounds (theorem 5.10), the one-sided distance test with a counterexample to its converse (proposition 5.12), and non-identifiability of the number of unmodeled sources (proposition 5.13).

(c) New section 10.6: analytic sieving as iterated conditioning, with the parity barrier as observational equivalence.

(d) New references [19, 20, 21, 22, 23, 27, 32, 41, 45]; new script `check_context_decomposition.py`; abstract, introduction, summary, open problems, conclusion and checks updated.

Version 7.

(a) New section 10.4: general (unsaturated) admissibility data, effective sets, the effective lifting criterion with first- and second-order certificates (theorem 10.13), the secondary obstruction and its indeterminacy interval (definition 10.14 and proposition 10.15), worked examples (example 10.16), the stratified obstruction types (definition 10.17), and the relation to classical obstruction theory (remark 10.18).

(b) The definition of an obstructed claim now covers extension-obstructed claims; abstract, introduction, summary, open problems, conclusion and checks updated; new script `check_obstruction_tower.py`.

Version 6.

(a) Section 13.6 rewritten as convex-order Beck comparisons: unit and multiplication comparisons with exact martingale couplings (proposition 13.8), the orientation obstruction (proposition 13.9), the comparability obstruction for (Cm) for both candidate laws (proposition 13.10), and corollary 13.11. The earlier suggestion that λ^σ might be a lax mixed distributive law in the convex order is withdrawn.

(b) New remark 13.12 on lifting orders, including the degeneracy of the naive liftability preorder; new "Structure of the argument" paragraph; abstract, summary, open problems and conclusion updated; new script `check_convex_order.py`.

Version 5. Relative to version 4 and to an intermediate draft that added two sections on 2-categorical lifting, BKP 2-monadicity and doctrinal adjunction:

(a) The intermediate sections are replaced by section 13. They worked in a 2-category of C^B-coalgebras whose 2-cells were described as natural transformations between unspecified "family actions"; no such structure had been defined, and on sets regarded as discrete categories the only 2-cells are identities. Sections 13.1 and 13.5 state the ambient 2-category and what it does and does not provide.

(b) The claimed cofree coalgebra used $\Delta^B(c) = (c, e_B)$ for a chosen e_B, which is not the comultiplication $(y, c) \mapsto ((y, c), c)$ of C^B; the claimed adjunction unit $x \mapsto (x, e_B)$ is not the unit of the cofree adjunction and is not natural. Both are corrected in proposition 13.2. The flexibility proof, which relied on them and used an undefined map κ_B in the role of the barycenter β_B, is removed, as are the strictification corollary, the 2-monadicity proposition (whose codescent and 2-cell arguments depended on the missing structure) and the doctrinal lax-structure formula.

(c) Retained and proved: the lifted monad (proposition 13.4), with the coalgebra category identified as $\mathbf{Set}/A_B$ (proposition 13.1), and the recovery of λ^B from the cofree coalgebra. New: bialgebras as convex spaces over A_B (proposition 13.5) and a first convex-order comparison of the shift law's defects (superseded in version 6 by section 13.6).

(d) References: the intermediate draft cited "Bourke and Lack, *Monads and theories*, Adv. Math. 287 (2015)"; *Monads and theories* is by Bourke and Garner, and two-dimensional monadicity is [12], now cited instead. A duplicate entry for [43] and an unused reference on weak bimonads are removed; broken cross-references are fixed.

(e) CausalIDView is now attributed to [34]; it is not gradient-free, and the gradient-free method applied to it is cited to [9, Section 51].

(f) Empirical corrections: the 70–71% coverage refers to the structured backbone's bootstrap intervals, and coverage of the generic-prior posterior has not been measured; on shallow-MLP worlds 0.104 is the generic-prior posterior and 0.103 the structured backbone; the 30 bootstrap refits apply to the backbone row only, while posterior rows come from pooled Metropolis chains.

Version 4.

(a) New section 10: admissibility data, admissible tower, lifting criterion, failure levels, obstruction certificates, pooling behavior, and the interpretation of obstructed claims.

(b) New reference [39]; abstract, introduction, summary, open problems, conclusion and computational checks updated.

Version 3.

(a) New section 9: base-relative tower with coherent contexts, canonical hierarchy section, and a strict mixed distributive law satisfying all four Beck axioms (theorem 9.8), natural in the base, with pooling and conditioning results.

(b) New remark 6.5 making explicit that $\mathsf{C}_3(\mathsf{D}X)$ contains D^4X, which the first version's $\lambda^{[3]}$ omitted; the absolute-tower laws are now labelled Option 1.

(c) Abstract, introduction, summary, open problems and conclusion rewritten around the design choice; repository link added.

Version 2 (relative to version 1).

1. Base category changed from finite sets to **Set**, since $\mathsf{D}(X)$ is infinite.
2. Hyp redefined as $\Theta \times \mathsf{D}(-)$; transport of directions and splines by $v_j \circ f$ removed (ill-typed, remark 3.4); the compatibility square is now naturality of q, not a definition.
3. Composition of posteriors normalized; distributivity and Cond/$\otimes$ commutation corrected (propositions 4.8 and 4.12); chain rule stated with its conditional likelihood.
4. Observational equivalence: the congruence for $\otimes$ needs total compatibility, not q-monoidality; the congruence for Cond claimed earlier is false and is replaced by proposition 5.5.
5. The comonad is now an instance of a general theorem (theorem 6.1); Sh_3 replaced by Sh; naturality of δ is proved.
6. J is defined as $\mathrm{Hyp}(j)$ so that CL1–CL3 are proved; the seventh tuple component of the earlier construction is no longer needed.
7. K/J coherence corrected to use $\mathsf{D}^3 j$ (it was ill-typed with $\mathsf{D}^2 j$).

8. $\lambda^{[3]}$ had four components instead of five and an ill-typed shadow; it is replaced by λ^{σ}. The earlier unit and comultiplication theorems were false (theorem 8.3). The averaging law $\bar{\lambda}$ and lemma 8.1 are new.

9. The weak distributive law and 2-cell α are withdrawn (remark 8.9). The contradiction between the claimed proof of hexagonal coherence and its listing as an open problem is resolved by removal.

10. Composite identifiability (which trivializes Hyp) removed; it is not needed. The "iff" proof corrected (theorem 12.3); condition (c) of nondegeneracy (finiteness) dropped as automatic.

11. Bibliography corrected. The earlier references "Böhm and Lack, *Weak distributive laws*, J. Algebra 282 (2004)" and "Lack, *Composing monads using semimonads*, J. Algebra 281 (2004)" could not be located and are removed; they are replaced by [44, 16, 17, 28, 29, 38, 15]. The de Finetti reference now cites its journal version [25].

12. Empirical claims toned down: no parity claim with TabPFN; BBP attribution removed; miscalibration flagged.

Items for the author to confirm.

13. Source and protocol of the TabPFN reference values in the benchmark table.

14. The nominal number of directions ($r = 3$ in the setup) versus the 6.6 planted directions in the spectral pre-pass.

15. Bootstrap standard errors for the chain-pooling curve.

16. Bibliographic details of the three self-citations [7, 8, 9], which are reproduced as given in the previous version.

17. Protocols for the logistic-regression and constant-predictor rows of the benchmark table.

18. The metric (Manski or IV RMSE) behind the shallow-MLP figures 0.104 and 0.103.

C Computational checks

All identities and counterexamples in sections 5 and 8, theorem 6.1 and proposition 7.2 were checked in exact rational arithmetic on random instances with $X = \{0, 1\}$, two structural descriptors, and maps to a three-element set (one injective, one not). The checks confirm the comonad laws and naturality of ε, δ; K/J coherence; naturality, (U), (Co) and (M) for $\bar{\lambda}$; the failure of (Cm) for $\bar{\lambda}$ on the explicit example of theorem 8.7; naturality and (Co) for λ^{σ} and the failures of (U), (M), (Cm); and the counterexample of proposition 5.5. These are in `check_tower.py`.

For the base-relative tower, `check_base_relative.py` checks, on random coherent and incoherent contexts over $B = \{0, 1\}$: the comonad laws; naturality of λ^{B} in Y; all four axioms (U), (Co), (M), (Cm); closure of coherent contexts under barycenter; affineness and coherence of σ_B; preservation of coherence and commutation with σ and λ^{B} under base change along an injective and a non-injective map; example 9.11; and proposition 9.12. All checks pass. `check_stochastic_basechange.py` checks proposition 9.14 (algebra morphism, coherence, σ, functoriality on the Kleisli category, agreement with proposition 9.13 for deterministic kernels, comonad morphism and law), corollary 9.15 (transport of saturated data and lifts, pullback of certificates, an obstruction that

vanishes, and non-increase of o_3 by linear programming, including merging kernels), proposition 9.16, and that level-one and level-two Kantorovich distances between transported contexts are bounded by $\max_b \mathrm{TV}(f(b), g(b))$. All checks pass. The scripts are in the repository cited in the code availability section.

For section 10, `check_lifting.py` checks the three examples of example 10.6 with their certificates; surjectivity $\mu(L_3) = L_2$ via explicit lifts; monotonicity of failure; the distance formula of proposition 10.9 and its affinity under pooling; the minimax identity of theorem 10.8 by solving the primal and dual linear programs; the cancellation example of proposition 10.10; and that canonical contexts lie in the committed tower. The linear programs are solved in floating point; all other checks are exact. All 17 checks pass.

For section 10.7, `check_support_floors.py` checks, in exact rational arithmetic on random credal polygons and kernels between bases of sizes two and three: proposition 10.23; proposition 10.24(ii) and (iii), with 93 strict cases of the policy inclusion and 132 strict oplax comparisons in 300 trials each; theorem 10.26(i)–(iv); and example 10.27. It also checks the floored gluing of the binary triangle and the floored hexagon used in [4, 6]. All checks pass.

For section 13, `check_coalgebras.py` checks that coalgebra structures are of the form $\langle \mathrm{id}, g \rangle$ with coassociativity automatic; the cofree adjunction bijection; that η and μ are maps over A_B for the lifted monad; the recovery of λ^B from the cofree coalgebra; the bialgebra condition for $(\mathrm{Sh}B, \mu, \sigma_B)$ and its failure for a non-affine context map; and the level-one convex-order inequalities on a family of convex test functions. All 10 checks pass.

For section 13.6, `check_convex_order.py` certifies, in exact arithmetic: the unit and multiplication comparisons of proposition 13.8 by constructing the martingale couplings and checking both marginals and the mean condition atom by atom, on random instances; the failure of the reverse inequalities, with the witness $\|v\|_2^2$; the resulting orientation obstruction of proposition 13.9; and, for both λ^σ and $\bar{\lambda}$, that the two sides of (Cm) in proposition 13.10 differ, have equal barycenters, and are incomparable, with the witnesses φ_1 and φ_2. All 14 checks pass.

For section 10.4, `check_obstruction_tower.py` verifies example 10.16 exactly: the fibres, o_3 along each fibre, the indeterminacy intervals, the explicit lift and both certificates, and the pooling example. It checks exactly, on random instances, that μ_B is 1-Lipschitz, that points of E_1 have the explicit lift of theorem 10.13, and that saturated data have $E_2 = L_2$. Using linear programs on random unsaturated data, it checks that $x_1 \in E_1$ iff $o_3^-(x_1) = 0$ and that $d(x_1, E_1) \le o_3^-(x_1)$; for points outside E_1, the dual certificate is rounded to rationals and verified to separate exactly. All 21 checks pass.

For sections 5.2, 5.3 and 10.6, `check_context_decomposition.py` checks: exact and unique recovery of weights for affinely independent sources; for affinely dependent sources, an exact dependency giving two decompositions and a nondegenerate LP interval; the exclusive-region formula and per-observation attribution on a worked example; for supports only, that the interval of proposition 5.8 is attained at both ends and cannot be extended, and, for three sources, that feasible allocations coincide with the core of the belief function; for contamination, exact primal–dual certificates bracketing $\mu_{\min}$, the interval $[\mu_{\min}, 1]$, the inequality $d \le 2\mu_{\min}$, the equivalence $\mu_{\min} = 0 \iff x \in \mathrm{conv}(A)$, the counterexample of proposition 5.12, and the merging of unmodeled sources; and, for $N = 3000$, iterated versus single conditioning, the bounds on prime factors of survivors, the primes as endpoint, and the pure parity of Selberg's weights (with the relative difference of their sieve data reported as an illustration). All 20 checks pass.

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